

Introduction to Cryptography

Lecture#1

Contents

- Objective of Course
- What is information security?
- What are security goals ?
- Overview on the field of cryptology
- Definition Cryptography
- Understand security
 - Attacks threatening security goals
 - Security attack – services and mechanism

Introduction

- We are living in the information age. We need to keep information about every aspect of our lives. In other words, information is an asset that has a value like any other asset. As an asset, information needs to be secured from attacks.
- To be secured, information needs to be hidden from unauthorized access (**confidentiality**), protected from unauthorized change (**integrity**), and available to an authorized entity when it is needed (**availability**).

What is Information Security ?

- Information Security is the process of protecting data from unauthorized access , use , disclosure , destruction, modification or disruption

Cryptography

- **Cryptography** is the practice of securing information by transforming it into an unreadable format so that only authorized parties can access it. It is used to ensure **confidentiality, integrity, authentication, and non-repudiation** of data.
- **Cryptography** is the study and practice of techniques for secure communication in the presence of third parties called adversaries الخصوم.

Cryptography

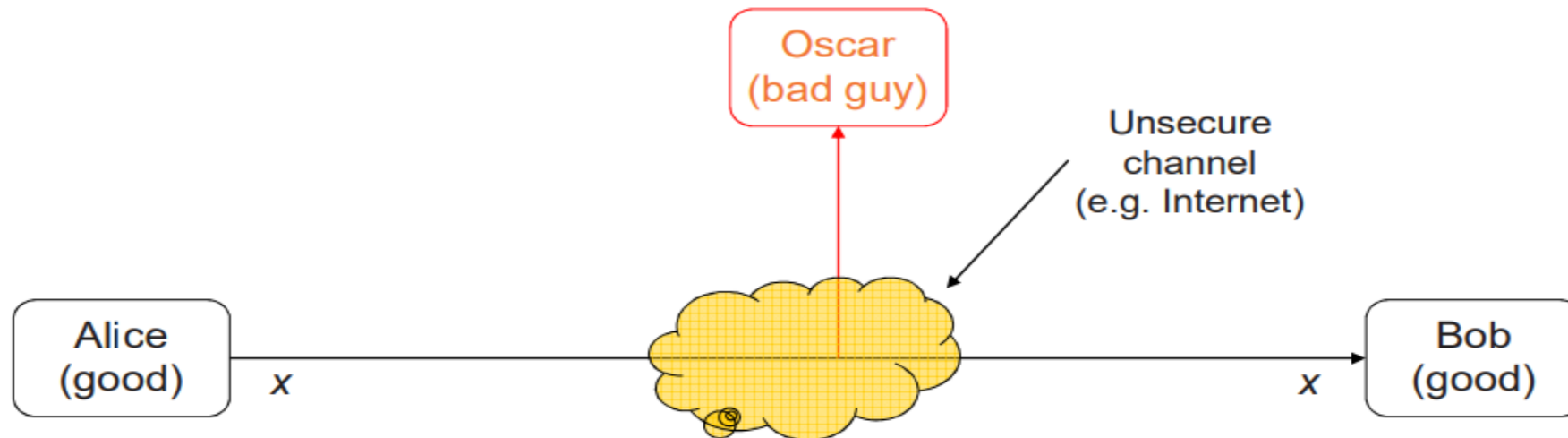
- Cryptography (from Greek *kryptós*, “hidden” and graphein, “writing”) is the traditionally, the study of means of converting information from its normal , comprehensible form into an incomprehensible format(unreadable) rendering it unreadable without secret knowledge- the art of encryption.
- The study of how to circumvent the use of cryptography is called Cryptanalysis or codebreaking.
- Cryptography and cryptanalysis are sometimes grouped under the umbrella term Cryptology encompassing the entire subject.

Cryptography

- Cryptography is the study of mathematical techniques related to aspects of information security such as confidentiality , and data integrity

Understand Security

- To understand the security , you need to understand what sort of attacks are possible



Understand Security

- A **Channel** is a means of conveying information from one entity to another.
- An **unsecured Channel** is one from which parties other than those for which the information is intended can reorder , delete , insert , or read .
- A **secure channel** is one from which an attacker does not have ability to reorder, delete , insert or read.

Understand Security

- **Attacker** is anyone who tries to defeat the information security services being provided between sender and receiver.
- Attacker=Adversary, enemy, opponent, tapper, eavesdropper, intruder.

SECURITY GOALS

- security goals: confidentiality, integrity, and availability



SECURITY GOALS -Confidentiality

- Ensuring that no one can read the message except the intended receiver
- Confidentiality is probably the most common aspect of information security. We need to protect our confidential information. An organization needs to guard against those malicious actions أنشطة خبيثة that endanger the confidentiality of its information. In the military, concealment إخفاء of sensitive information is the major concern. In industry, hiding some information from competitors is crucial to the operation of the organization. In banking, customers' accounts need to be kept secret

SECURITY GOALS - Integrity

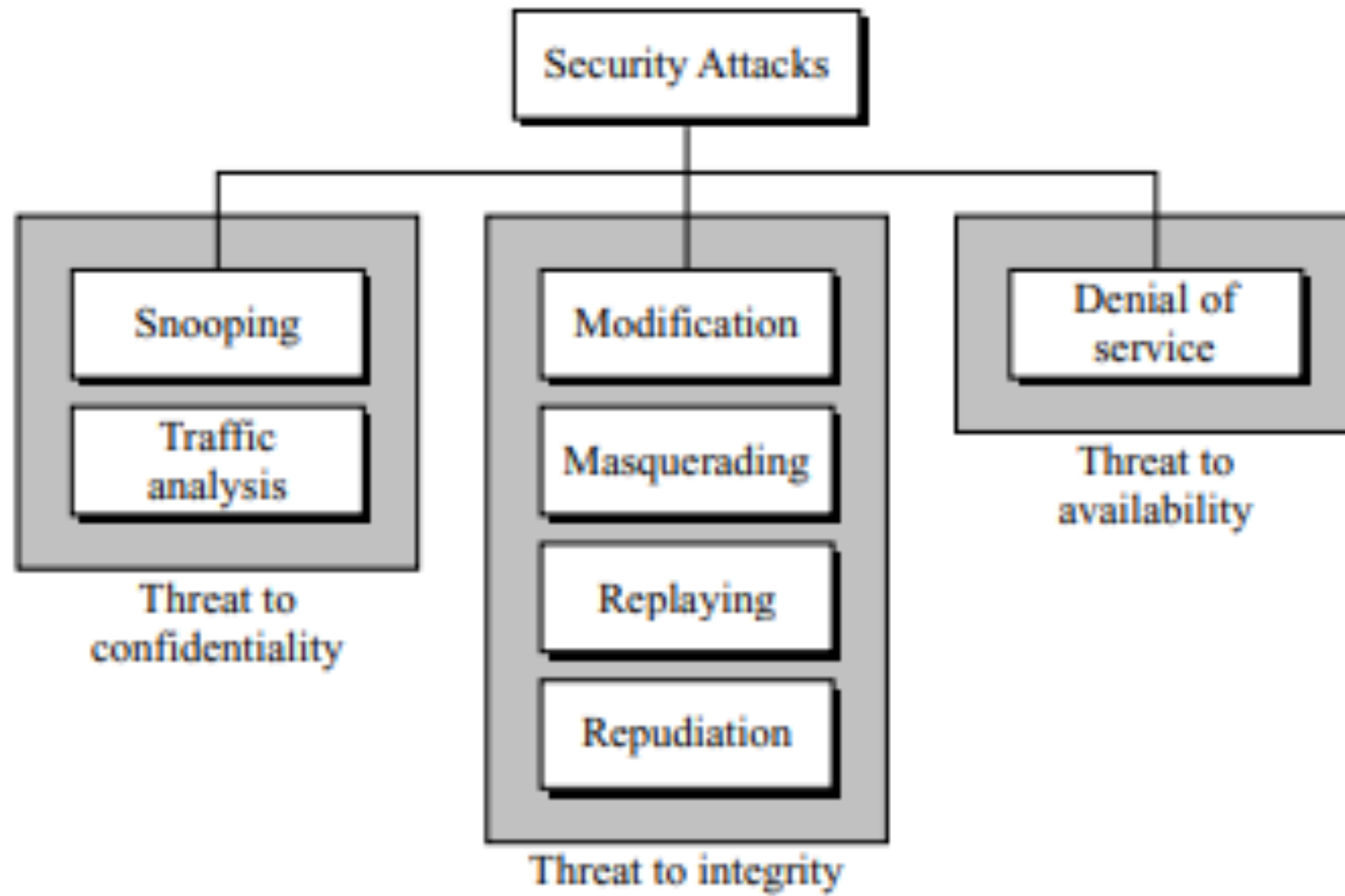
- Assuring the receiver that the received message has not been altered in any way from the original.
- Information needs to be changed constantly. In a bank, when a customer deposits or withdraws money, the balance of her account needs to be changed. Integrity means that **changes need to be done only by authorized entities** and through authorized mechanisms. Integrity violation is not necessarily the result of a malicious act; an interruption in the system, such as a power surge, may also create unwanted changes in some information.

SECURITY GOALS -Availability

- The information created and stored by an organization needs to be available to authorized entities. **Information is useless if it is not available.** Information needs to be constantly changed, which means it must be accessible to authorized entities. The unavailability of information is just as harmful for an organization as the lack of confidentiality or integrity. Imagine what would happen to a bank if the customers could not access their accounts for transactions

ATTACKS

- Our three goals of security confidentiality, integrity, and availability can be threatened by security attacks. Although the literature uses different approaches to categorizing the attacks, we will first divide them into three groups related to the security goals. Later, we will divide them into two broad categories based on their effects on the system. Figure 1.2 shows the first taxonomy



Taxonomy of attacks with relation to security goals

Attacks Threatening Confidentiality

هجمات تهدد السرية

- In general, two types of attacks threaten the confidentiality of information:

- Snooping :التطفل : release of message contents

Snooping refers to unauthorized access to or interception of data اعتراض البيانات.

- Traffic analysis.: is the analysis of patterns of communication in order to learn secret information

Attacks Threatening Integrity

- The integrity of data can be threatened by several kinds of attacks:
 - Modification التعديل
 - Masquerading التجميع الجماعي
 - Replaying إعادة التشغيل
 - Repudiation. الرفض

Attacks Threatening Integrity– Modification

(after intercepting or accessing data , attacker modifies it)

- After intercepting or accessing information, the attacker modifies the information to make it beneficial to herself.
- For example,
- a customer sends a message to a bank to do some transaction. The attacker intercepts the message and changes the type of transaction to benefit herself.

Attacks Threatening Integrity- Masquerading

(The attacker impersonate the sender or receiver)

-
- Masquerading, or spoofing التكرار أو الخداع, happens when the attacker impersonates ينتحل شخصية شخص ما somebody else.
 - For example,
 - an attacker might steal the bank card and PIN of a bank customer and pretend يدعى that she is that customer. Sometimes the attacker pretends instead to be the receiver entity. For example, a user tries to contact a bank, but another site pretends that it is the bank and obtains some information from the user.

Attacks Threatening Integrity- Replaying

(obtain a copy of data sent by a sender and later tries to replay it)

- The attacker obtains a copy of a message sent by a user and later tries to replay it.
- For example, a person sends a request to her bank to ask for payment to the attacker, who has done a job for her. The attacker intercepts the message and sends it again to receive another payment from the bank.

Attacks Threatening Integrity – Repudiation

(the sender of the message might later deny that he sent the message)

- This type of attack is different from others because it is performed by one of the two parties in the communication: the sender or the receiver.
- The sender of the message might later deny that she has sent the message; the receiver of the message might later deny that he has received the message.
- An example of denial by the sender would be a bank customer asking her bank to send some money to a third party but later denying that she has made such a request.
- An example of denial by the receiver could occur when a person buys a product from a manufacturer and pays for it electronically, but the manufacturer الشركة المصنعة later denies having received the payment and asks to be paid.

Attacks Threatening Availability

- **Denial of Service** (DoS) is a very common attack. It may slow down or totally interrupt the service of a system.
- Example :
- The attacker might intercept and delete a server's response to a client, making the client to believe that the server is not responding
- The attacker may also intercept requests from the client, causing the clients to send requests many times and overload the system

Types of Attacker

- **Passive Attacker**

- ☐ Just obtain information
- ☐ Difficult to detect

- **Active Attacker**

- ☐ Obtain and may change the information
- ☐ Easier to detect than to prevent

Passive Versus Active Attacks

Table 1.1 *Categorization of passive and active attacks*

<i>Attacks</i>	<i>Passive/Active</i>	<i>Threatening</i>
Snooping Traffic analysis	Passive	Confidentiality
Modification Masquerading Replaying Repudiation	Active	Integrity
Denial of service	Active	Availability

Passive Attacks

- In a passive attack, the attacker's goal is just to obtain information. This means that the attack does not modify data or harm the system. The system continues with its normal operation. However, the attack may harm the sender or the receiver of the message. Attacks that threaten confidentiality□ snooping and traffic analysis□ are passive attacks. The revealing of the information may harm the sender or receiver of the message, but the system is not affected. For this reason, it is difficult to detect this type of attack until the sender or receiver finds out about the leaking of confidential information. Passive attacks, however, can be prevented by encipherment of the data.

Active Attacks

- An active attack may change the data or harm the system. Attacks that threaten the integrity and availability are active attacks. Active attacks are normally easier to detect than to prevent, because an attacker can launch them in a variety of ways.

Security Attacks, Mechanisms and Services

- **Security Attacks :**

Any action that compromises the security of information.

ای عمل من شأنه المساس بامن المعلومات

- **Security Mechanisms:-**

A mechanism that is designed to detect, prevent, or recover from security attack.

- **Security Services**

A service that enhances the security of data processing systems and information transfer . A security service makes use of one or more security mechanism

TECHNIQUES

- Two techniques are prevalent today: one is very general (cryptography) and one is specific (steganography).

Principles of modern-day cryptography.

- Data Confidentiality سرية البيانات
- Data Integrity تكامل البيانات
- Authentication المصادقة
- Non-repudiation of data عدم الانكار

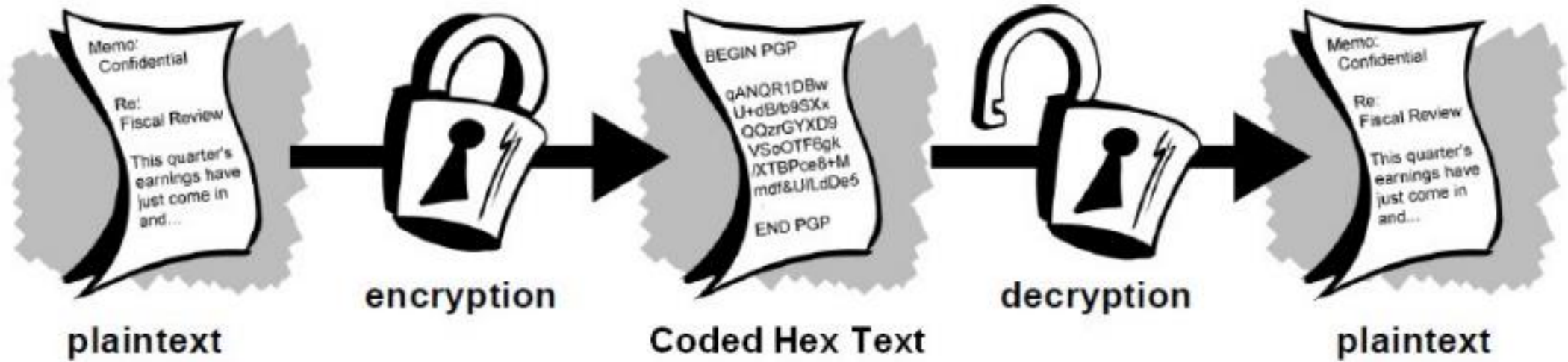
Definition Cryptography

- Cryptography is the science of using mathematics to encrypt and decrypt data. (Phil Zimmermann)
- Cryptography is the art and science of keeping messages secure. (Bruce Schneier)
- The art and science of concealing إخفاء the messages to introduce secrecy in information security is recognized as cryptography.



Terminologies

- **Plaintext:** A message is plaintext النص العادي (sometimes called cleartext).
- **Encryption :** The process of disguising a message in such a way as to hide its substance is encryption. تمويه
- **Decryption:** An encrypted message is ciphertext. The process of turning ciphertext back into plaintext is decryption.
- **A cipher** (or cypher) is an algorithm for performing encryption or decryption—a series of well-defined steps that can be followed as a procedure.



Terminologies

- A cryptosystem **نظام التشفير** is an implementation of cryptographic techniques and their accompanying infrastructure **البنية التحتية المصاحبة** to provide information security services **خدمات الامن**.
- A cryptosystem is also referred to as a cipher system. The various components of a basic cryptosystem are as follows
 - **Plaintext**
 - **Encryption Algorithm**
 - **Ciphertext**
 - **Decryption Algorithm**
 - **Encryption Key**
 - **Decryption Key**

Terminologies

- While cryptography is the science of securing data, **cryptanalysis** تحليل الشفرات is the science of analyzing and breaking secure communication.
- **Classical cryptanalysis** involves an interesting combination of analytical reasoning, application of mathematical tools, pattern finding, patience, determination, and luck.
- **Cryptanalysts** محلل الشفرات are also called **attackers**.
- **Cryptology** embraces both cryptography and cryptanalysis.

History of Cryptography

- As civilizations evolved, human beings got organized in tribes قبائل, groups, and kingdoms مملكة. This led to the emergence ظهور of ideas such as power, battles معارك السلطة, supremacy التفوق, and politics سياسة. These ideas further fueled the natural need of people to communicate secretly with selective recipient which in turn ensured يضمن the continuous evolution of cryptography as well.
- The roots of cryptography are found in Roman and Egyptian civilizations الحضارة الرومانية والمصرية.

History of Cryptography

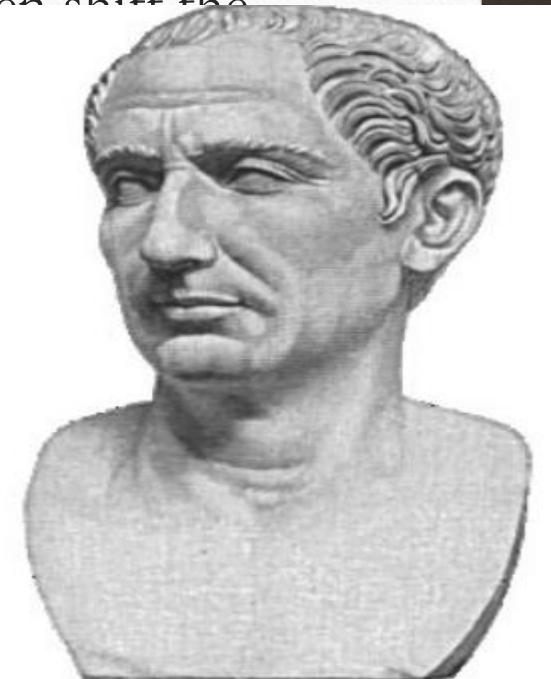
- Hieroglyph

The first known evidence دليل of cryptography can be traced to the use of 'hieroglyph'. Some 4000 years ago, the Egyptians used to communicate by messages written in hieroglyph.

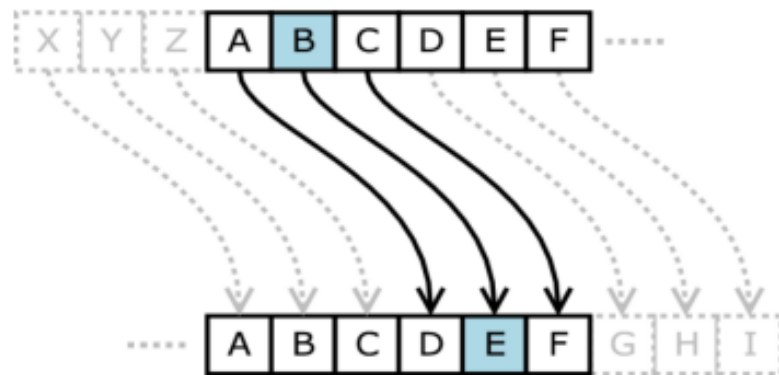


History of Cryptography

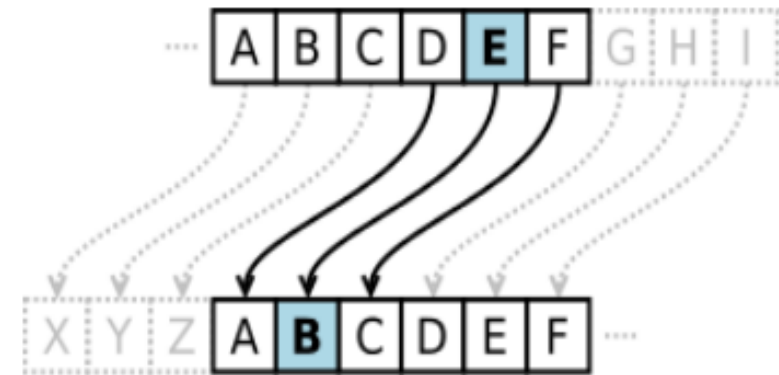
- Caesar Shift Cipher
- Caesar Shift Cipher, relies on shifting the letters of a message by an agreed number (three was a common choice), the recipient of this message would then shift the letters back by the same number and obtain the original message.
- The Caesar cipher is named after Julius Caesar ,
- who used it with a shift of three to protect messages of military significance.



Caesar Shift Cipher



Encryption



Decryption

PLAINTEXT : internet society ghana chapter

CYPHERTEXT : lqwhuqhw vrflhwb jkdqd fkdswhu

History of Cryptography - Kamasutra Cipher

- The Kamasutra cipher is one of the earliest known substitution methods. It is described in the Kamasutra around 400 BC.
- The purpose was to teach women how to hide secret messages from prying eyes عيون المتطفلين .

History of Cryptography

Kamasutra Cipher

The techniques involves randomly pairing letters of the alphabet, and then substituting each letter in the original message with its partner.

UPPER HALF	W	Z	V	P	O	F	D	E	A	B	R	M	Y
LOWER HALF	N	H	G	X	K	S	I	C	J	U	T	Q	L

The key is the permutation of the alphabet.

INTERNET SOCIETY GHANA CHAPTER

DWRCTWCR FKEDCRL VZJWJ EZJXRCT

Goals and Services of Cryptography

- **Goal:**

The primary goal of cryptography is to secure important data on the hard disk or as it passes through a medium that may not be secure itself. Usually, that medium is a computer network.

- **Services:-**

Cryptography can provide the following services:

- Confidentiality (secrecy) السرية
- Integrity (anti-tampering) السلامة
- Authentication المصادقية
- Non-repudiation. عدم الانكار

Goals and Services of Cryptography

Confidentiality (secrecy) السرية

- Ensuring that no one can read the message except the intended receiver
- Data is kept secret from those without the proper credentials اوراق اعتماد, even if that data travels through an insecure medium وسيلة غير آمنة.

Integrity (anti-tampering) السلامة

- Assuring the receiver that the received message has not been altered in any way from the original.

طمانة المستقبل بان الرسالة المستلمة لم تتبدل

Goals and Services of Cryptography

Authentication المصادقية

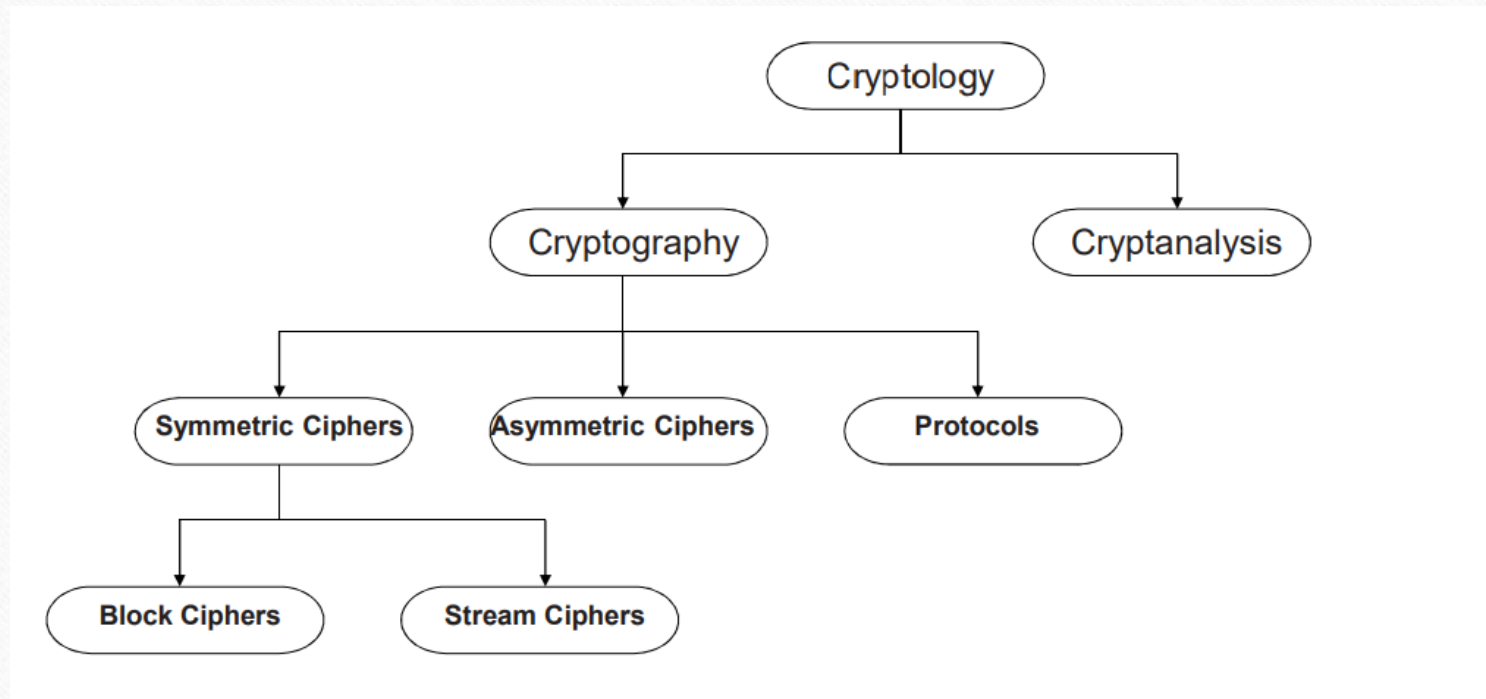
- Cryptography can help establish identity for authentication purposes The process of proving one's identity.

يمكن ان يساعد التشفير في اثبات الهوية لاغراض المصادقية

Non-repudiation

- A mechanism to prove that the sender really sent this message
- الية ان المرسل ارسل هذه الرسالة بالفعل

Classification of the Field of Cryptology



Some Basic Facts

- Ancient Crypto: Early signs of encryption in Egypt in ca. 2000 B.C. Letter-based encryption schemes (e.g., Caesar cipher) popular ever since.
- Symmetric ciphers: All encryption schemes from ancient times until 1976 were symmetric ones.
- Asymmetric ciphers: In 1976 public-key (or asymmetric) cryptography was openly proposed by Diffie, Hellman and Merkle.
- Hybrid Schemes: The majority of today's protocols are hybrid schemes, i.e., the use both
 - symmetric ciphers (e.g., for encryption and message authentication) and
 - asymmetric ciphers (e.g., for key exchange and digital signature).

Types of Cryptography

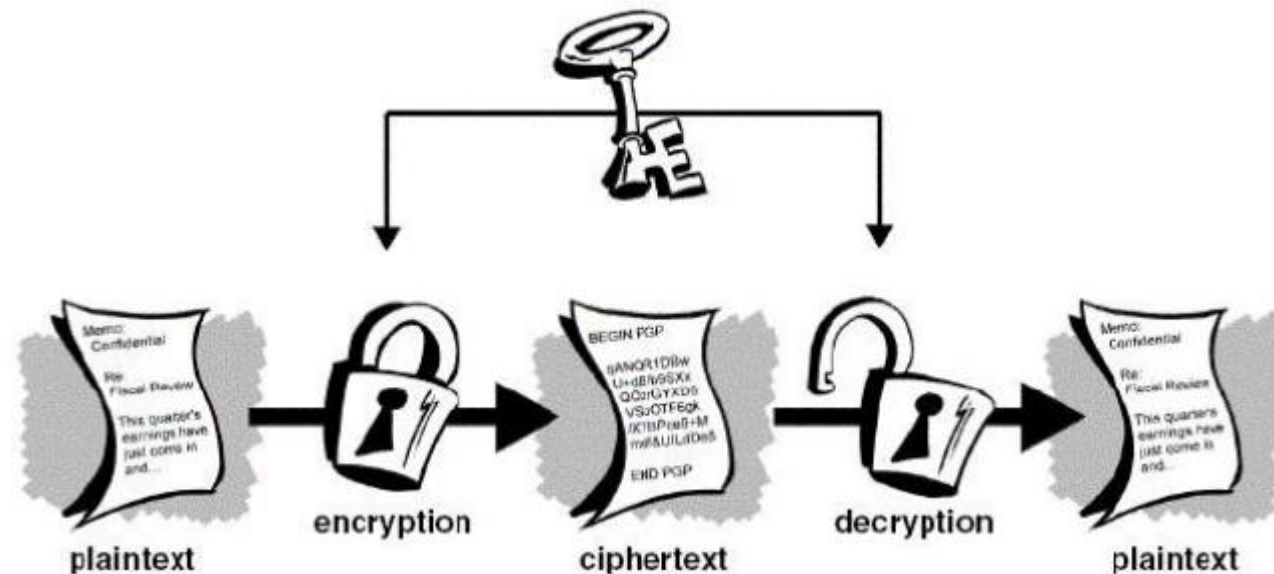
- Symmetric Key Cryptography التشفير المتماثل
- Asymmetric Key Cryptography التشفير غير المتماثل
- Hash Functions

Symmetric Key Cryptography

- Also known as Secret Key Cryptography or Conventional Cryptography, Symmetric Key Cryptography is an encryption system in which the sender and receiver of a message share a single, common key that is used to encrypt and decrypt the message.
- The Algorithm use is also known as a secret key algorithm or sometimes called a symmetric algorithm A key is a piece of information (a parameter) that determines the functional output of a cryptographic algorithm or cipher.

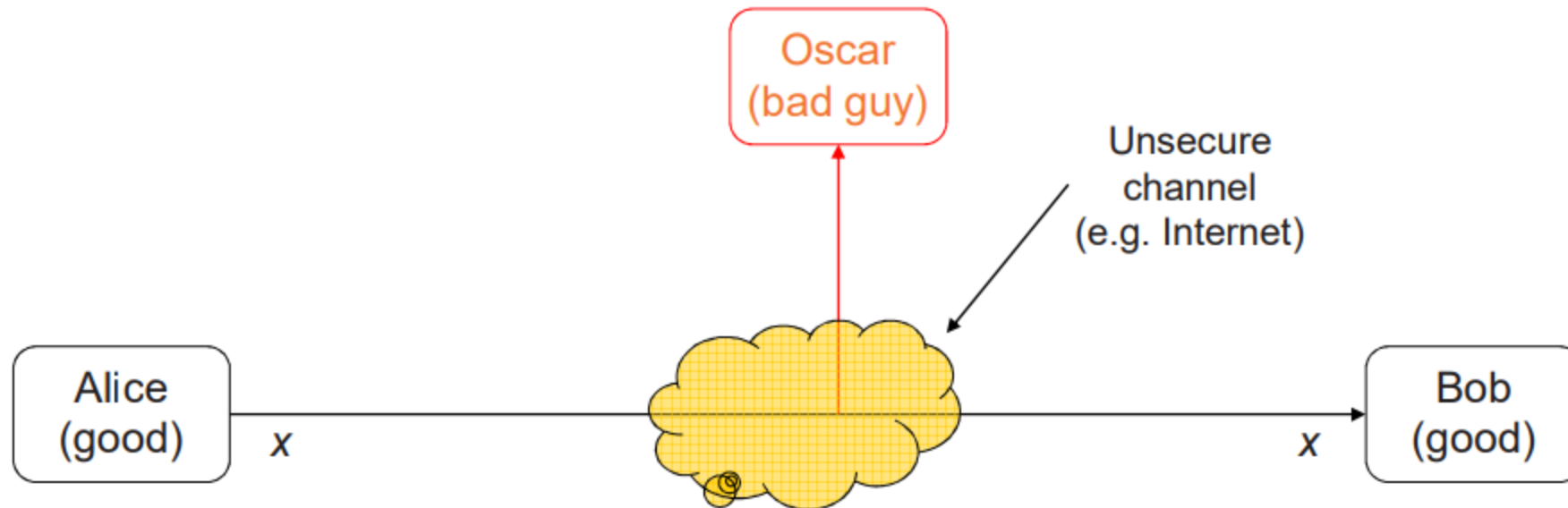
Symmetric Key Cryptography cont.,

- The key for encrypting and decrypting the file had to be known to all the recipients **المستلمين**. Else, the message could not be decrypted by conventional means.



Symmetric Cryptography

- Alternative names: private-key, single-key or secret-key cryptography.



Symmetric Cryptography

- Problem Statement:

1) Alice and Bob would like to communicate via an unsecure channel (e.g., WLAN or Internet).

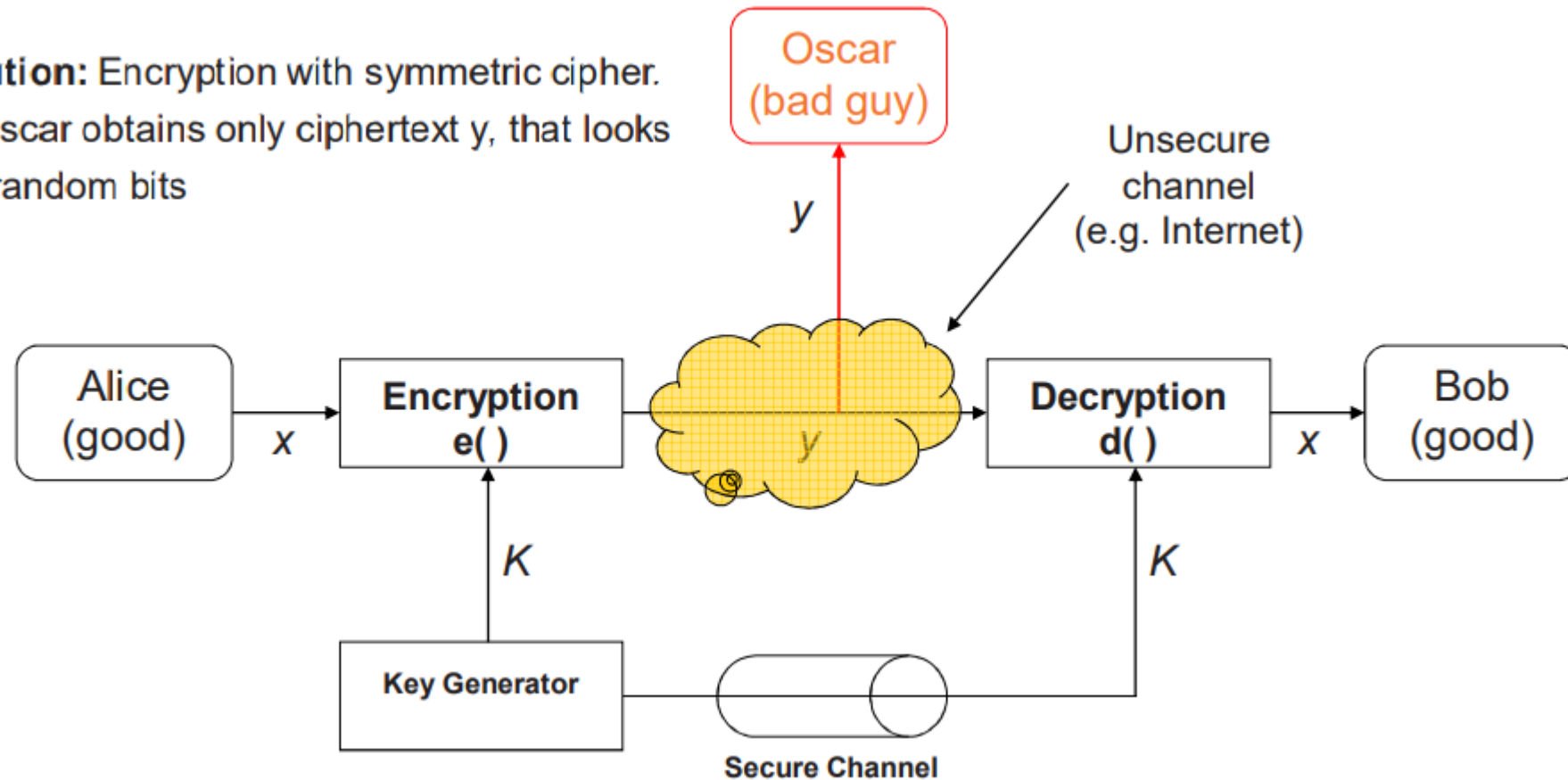
2) A malicious third party Oscar (the bad guy) has channel access but should not be able to understand the communication.

Symmetric Cryptography

- Solution:

Encryption with symmetric cipher. $\Rightarrow$ Oscar obtains only ciphertext y , that looks like random bits

Solution: Encryption with symmetric cipher.
⇒ Oscar obtains only ciphertext y , that looks like random bits



- x is the **plaintext**
- y is the **ciphertext**
- K is the **key**
- Set of all keys $\{K_1, K_2, \dots, K_n\}$ is the **key space**

Symmetric Cryptography

- Encryption equation $y = e_K(x)$
- Decryption equation $x = d_K(y)$

- Encryption and decryption are inverse operations if the same key K is used on both sides:

$$d_K(y) = d_K(e_K(x)) = x$$

- Important: The key must be transmitted via a **secure channel** between Alice and Bob.
- The secure channel can be realized, e.g., by manually installing the key for the Wi-Fi Protected Access (WPA) protocol or a human courier.
- However, the system is only secure if an attacker does not learn the key K !

⇒ **The problem of secure communication is reduced to secure transmission and storage of the key K .**

Examples - Symmetric Cryptography

Data Encryption Standard (DES)

- The Data Encryption Standard was published in 1977 by the US National Bureau of Standards. DES uses a 56 bit key and maps a 64 bit input block of plaintext onto a 64 bit output block of ciphertext. 56 bits is a rather small key for today's computing power.

Triple DES

- Triple DES was the answer to many of the shortcomings عيوب of DES. Since it is based on the DES algorithm, it is very easy to modify existing software to use Triple DES. It also has the advantage of proven reliability and a longer key length that eliminates يقضى على many of the shortcut attacks هجمات that can be used to reduce the amount of time it takes to break DES.

Examples - Symmetric Cryptography

Advanced Encryption Standard (AES) (RFC3602)

- Advanced Encryption Standard (AES) is an encryption standard adopted by the U.S. government. The standard comprises three block ciphers, AES-128, AES-192 and AES-256, adopted from a larger collection originally published as Rijndael.
- Each AES cipher has a 128-bit block size, with key sizes of 128, 192 and 256 bits, respectively. The AES ciphers have been analyzed extensively and are now used worldwide, as was the case with its predecessor, the Data Encryption Standard (DES).

Examples - Symmetric Cryptography

IDEA:-

- The International Data Encryption Algorithm was developed in 1991. It uses a 128 bit key to encrypt a 64 bit block of plaintext into a 64 bit block of ciphertext.
- IDEA's general structure is very similar to DES, it performs 17 rounds, each round taking 64 bits of input to produce a 64 bit output, using per-round keys generated from the 128 bit key.

Examples - Symmetric Cryptography

- Symmetric Key Cryptography - Examples Types of Cryptography
- Lucifer
- Madryga
- FEAL
- REDOC
- LOKI
- GOST
- CAST
- Blowfish Safer
- Crab RC5

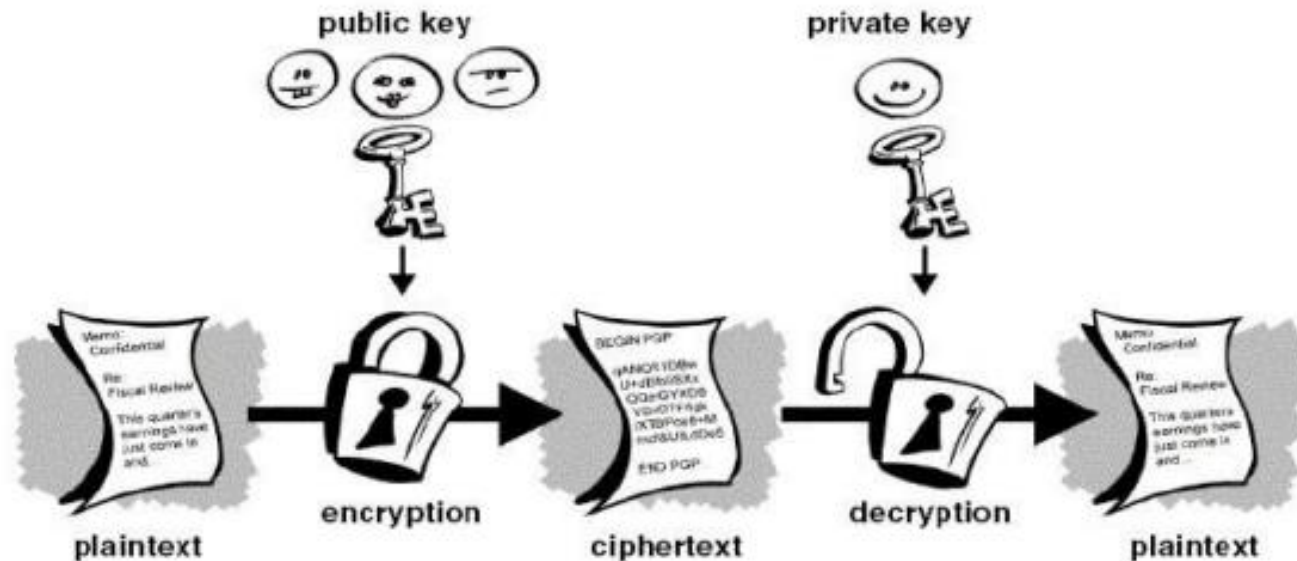
Problems with Conventional Cryptography

- Key Management
- Symmetric-key systems are simpler and faster; their main drawback is that the two parties طرفين must somehow exchange the key in a secure way and keep it secure after that.
- Key Management caused nightmare كابوس for the parties using the symmetric key cryptography. They were worried about how to get the keys safely and securely across to all users so that the decryption of the message would be possible.
- This gave the chance for third parties to intercept the keys in transit to decode the top-secret messages. Thus, if the key was compromised مساومة, the entire coding system was compromised and a “Secret” would no longer remain a “Secret”.
- This is why the “Public Key Cryptography” came into existence.

Asymmetric Key Cryptography

- Asymmetric cryptography, also known as **Public-key cryptography**, refers to a cryptographic algorithm which requires two separate keys, one of which is **private** and one of which is **public**. The public key is used to encrypt the

n



Step 1: Give your public key to the sender



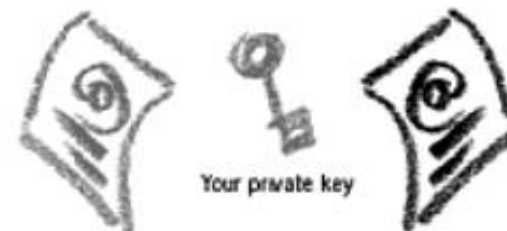
Step 2: Sender uses your public key to encrypt the plaintext



Step 3: Sender gives the ciphertext to you



Step 4: Use your private key (and passphrase) to decrypt the ciphertext



Asymmetric Key Cryptography

- Public Key Cryptography is a very advanced form of cryptography. Officially, it was invented by Whitfield Diffie and Martin Hellman in 1975.
- The basic technique of public key cryptography was first discovered in 1973 by the British Clifford Cocks of Communications-Electronics Security Group (CESG) of (Government Communications Headquarters - GCHQ) but this was a secret until 1997.

Asymmetric Key Cryptography - Examples

- Digital Signature Standard (DSS)
- Digital Signature Standard (DSS) is the digital signature algorithm (DSA) developed by the U.S. National Security Agency (NSA) to generate a digital signature for the authentication of electronic documents. DSS was put forth by the National Institute of Standards and Technology (NIST) in 1994, and has become the United States government standard for authentication of electronic documents. DSS is specified in Federal Information Processing Standard (FIPS) 186. D

Asymmetric Key Cryptography - Examples

- Algorithm – RSA
- RSA (Rivest, Shamir and Adleman who first publicly described it in 1977) is an algorithm for public-key cryptography. It is the first algorithm known to be suitable for signing as well as encryption, and one of the first great advances in public key cryptography.
- RSA is widely used in electronic commerce protocols, and is believed to be secure given sufficiently long keys and the use of up-to-date implementations.

- RSA Cryptanalysis

Rivest, Shamir, and Adelman placed a challenge in Martin Gardner's column in Scientific American (journal) in which the readers were invited to crack.

C=114,381,625,757,888,867,669,235,779,976,146,612,010,218,296,721,242,362,562,561,842,935,706,935
,245,733,897,830,597,123,563,958,705,058,989,075,147,599,290,026,879,543,541

-
- RSA Cryptanalysis
 - This was solved in April 26, 1994, cracked by an international effort via the internet with the use of 1600 workstations, mainframes, and supercomputers attacked the number for eight months before finding its Public key and its private key.
 - Encryption key = 9007
 - The message "first solver wins one hundred dollars".
 - Of course, the RSA algorithm is safe, as it would be incredibly difficult to gather up such international participation to commit malicious acts

-
- ElGamal
 - ElGamal is a public key method that is used in both encryption and digital signing.
 - The encryption algorithm is similar in nature to the Diffie-Hellman key agreement protocol
 - It is used in many applications and uses discrete logarithms.
 - ElGamal encryption is used in the free GNU Privacy Guard software

Why do we need Cryptanalysis?

- There is no mathematical proof of security for any practical cipher

- The only way to ensure security is to use a cryptosystem that is based on a mathematical problem that is hard to solve (a **Kerckhoff's Principle** is paramount in modern cryptography:

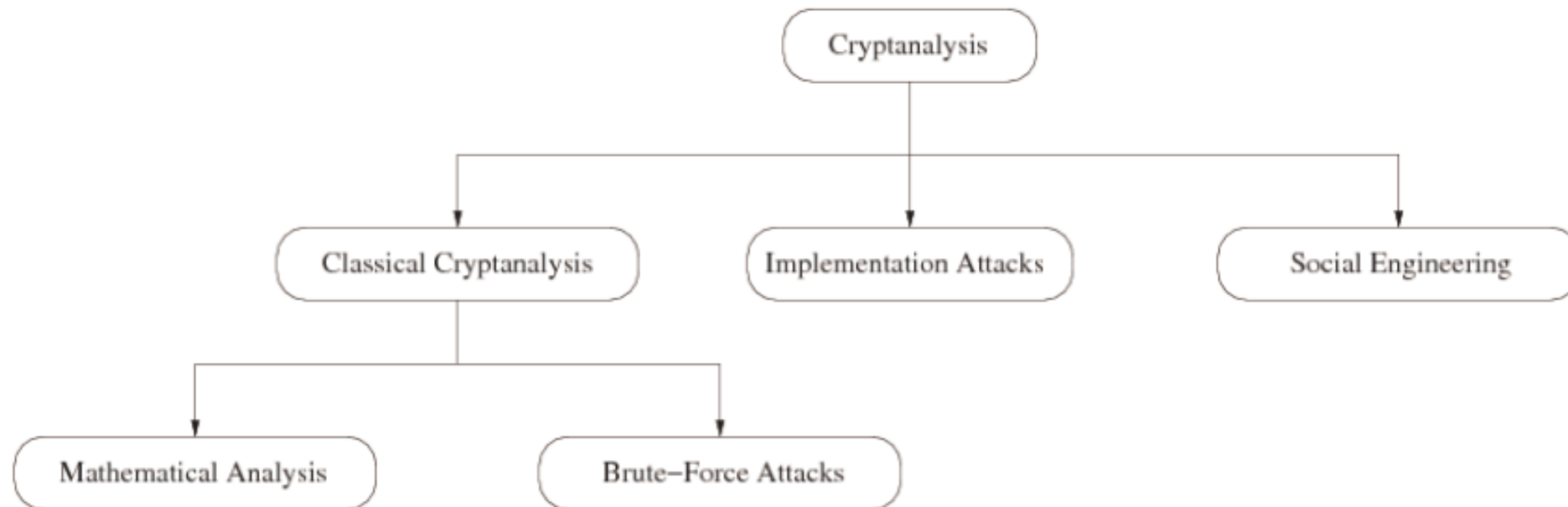
A cryptosystem should be secure even if the attacker (Oscar) knows all details about the system, with the exception of the secret key.

In order to achieve Kerckhoff's Principle in practice:

Only use widely known ciphers that have been cryptanalyzed for several years by good cryptographers! (*Understanding Cryptography* only treats such ciphers)

Remark: It is tempting to assume that a cipher is „more secure“ if its details are kept secret. However, history has shown time and again that secret ciphers can almost always be broken once they have been reversed engineered. (Example: Content Scrambling System (CSS) for DVD content protection.)

Cryptanalysis: Attacking Cryptosystems



Cryptography Techniques

Lecture#2

Difference between Symmetric-key and Asymmetric-key

- Encryption is a process to change the form of any message in order to protect it from reading by anyone.
- In **Symmetric-key encryption** the message is encrypted by using a key and the same key is used to decrypt the message which makes it easy to use but less secure. It also requires a safe method to transfer the key from one party to another.
- **Asymmetric Key Encryption:** is based on public and private key encryption techniques. It uses two different key to encrypt and decrypt the message. It is more secure than the symmetric key encryption technique but is much slower.

Symmetric Key Encryption

It only requires a single key for both encryption and decryption.

The size of cipher text is the same or smaller than the original plain text.

The encryption process is very fast.

It is used when a large amount of data is required to transfer.

Asymmetric Key Encryption

It requires two keys, a public key and a private key, one to encrypt and the other one to decrypt.

The size of cipher text is the same or larger than the original plain text.

The encryption process is slow.

It is used to transfer small amounts of data.

It only provides confidentiality.

It provides confidentiality, authenticity, and non-repudiation.

The length of key used is 128 or 256 bits

The length of key used is 2048 or higher

In symmetric key encryption, resource utilization is low as compared to asymmetric key encryption.

In asymmetric key encryption, resource utilization is high.

It is efficient as it is used for handling large amount of data.

It is comparatively less efficient as it can handle a small amount of data.

Security is less as only one key is used for both encryption and decryption purpose.

The Mathematical Representation is as follows-

$$P = D(K, E(K, P))$$

where $K \rightarrow$ encryption and decryption key

$P \rightarrow$ plain text

$D \rightarrow$ Decryption

$E(K, P) \rightarrow$ Encryption of plain text using K

It is more secure as two keys are used here- one for encryption and the other for decryption.

The Mathematical Representation is as follows-

$$P = D(K_d, E(K_e, P))$$

where $K_e \rightarrow$ encryption key

$K_d \rightarrow$ decryption key

$D \rightarrow$ Decryption

$E(K_e, P) \rightarrow$ Encryption of plain text using encryption key K_e . $P \rightarrow$ plain text

Symmetric-key encryption

- **Definition** Consider an encryption scheme consisting of the sets of encryption and decryption transformations $\{E_e : e \in K\}$ and $\{D_d : d \in K\}$, respectively, where K is the key space.
- The encryption scheme is said to be **symmetric-key** if for each associated encryption/decryption key pair (e, d) , it is computationally “easy” to determine d knowing only e , and to determine e from d .
- Since $e = d$ in most practical symmetric-key encryption schemes, the term symmetric key becomes appropriate. Other terms used in the literature are **single-key, one-key, private key, and conventional encryption.**

Example : illustrates the idea of symmetric-key encryption.

Example (*symmetric-key encryption*) Let $\mathcal{A} = \{A, B, C, \dots, X, Y, Z\}$ be the English alphabet. Let $\mathcal{M}$ and $\mathcal{C}$ be the set of all strings of length five over $\mathcal{A}$. The key e is chosen to be a permutation on $\mathcal{A}$. To encrypt, an English message is broken up into groups each having five letters (with appropriate padding if the length of the message is not a multiple of five) and a permutation e is applied to each letter one at a time. To decrypt, the inverse permutation $d = e^{-1}$ is applied to each letter of the ciphertext. For instance, suppose that the key e is chosen to be the permutation which maps each letter to the one which is three positions to its right, as shown below

$$e = \begin{pmatrix} A & B & C & D & E & F & G & H & I & J & K & L & M & N & O & P & Q & R & S & T & U & V & W & X & Y & Z \\ D & E & F & G & H & I & J & K & L & M & N & O & P & Q & R & S & T & U & V & W & X & Y & Z & A & B & C \end{pmatrix}$$

Example

A message

$m =$ THISC IPHER ISCER TAINL YNOTS ECURE

is encrypted to

$c = E_e(m) =$ WKLVF LSKHU LVFHU WDLQO BQRWV HFXUH.

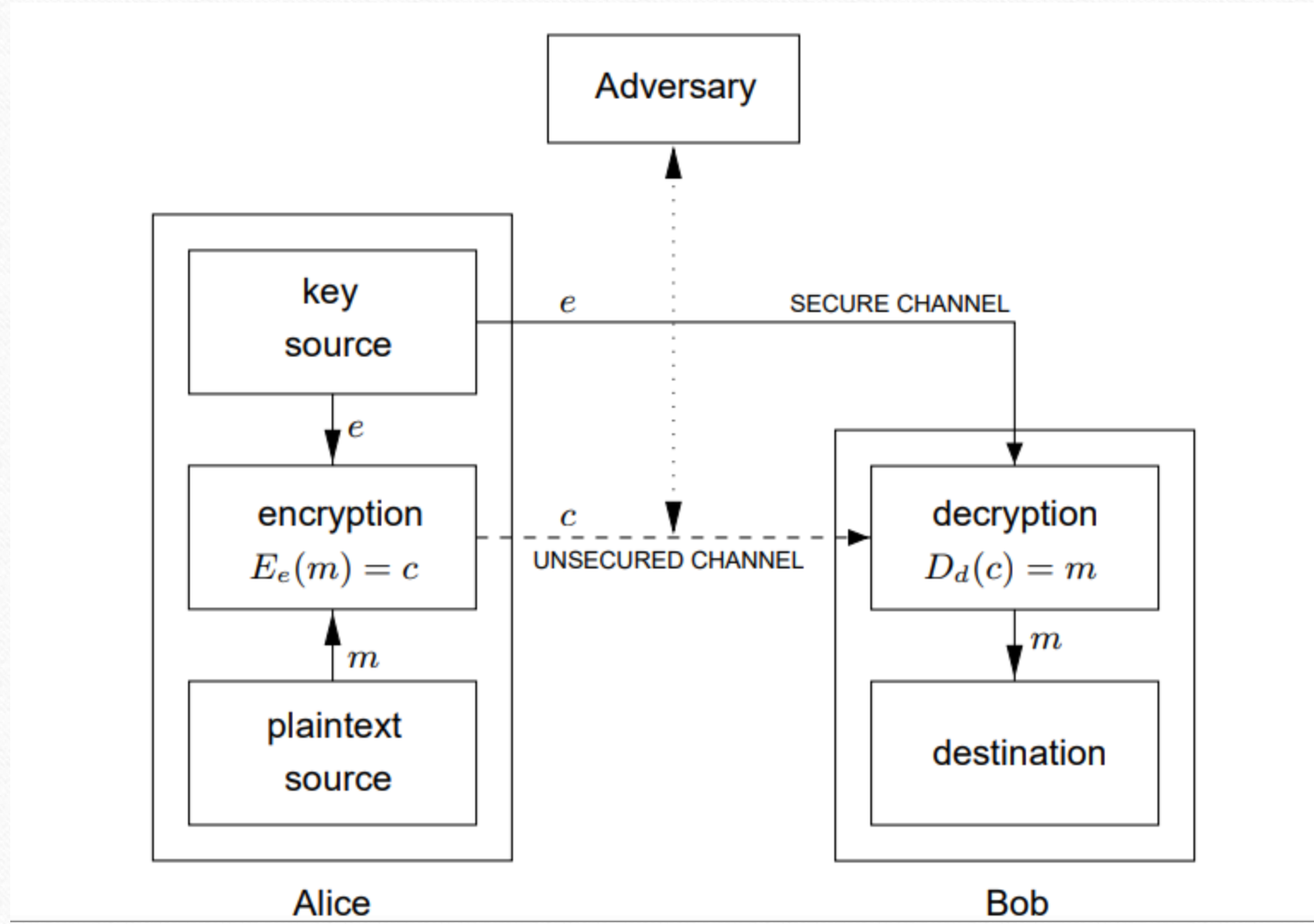


Figure 1.7: Two-party communication using encryption, with a secure channel for key exchange. The decryption key d can be efficiently computed from the encryption key e .

It is assumed that all parties know the set of encryption/decryption transformations (i.e., they all know the encryption scheme). As has been emphasized several times the only information which should be required to be kept secret is the key d . However, in symmetric-key encryption, this means that the key e must also be kept secret, as d can be deduced from e . In Figure 1.7 the encryption key e is transported from one entity to the other with the understanding that both can construct the decryption key d .

There are two classes of symmetric-key encryption schemes which are commonly distinguished: *block ciphers* and *stream ciphers*.

Types of Symmetric-key

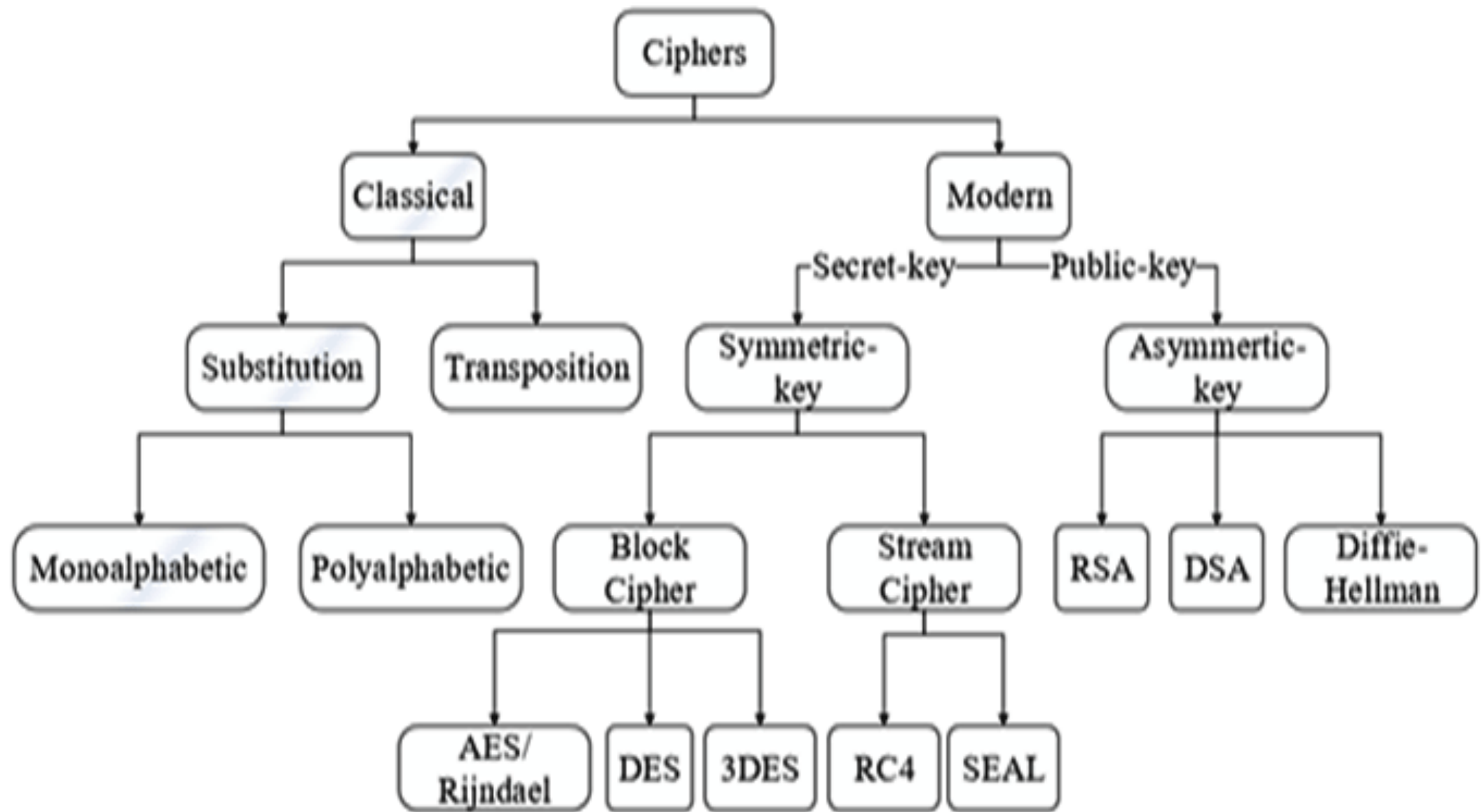
- There are two classes of symmetric-key encryption schemes which are commonly distinguished:
 - (1) Block ciphers
 - (2) Stream ciphers

Block Cipher

Definition A *block cipher* is an encryption scheme which breaks up the plaintext messages to be transmitted into strings (called *blocks*) of a fixed length t over an alphabet $\mathcal{A}$, and encrypts one block at a time.

Two important classes of block ciphers are:-

- 1- Substitution ciphers
- 2- Transposition cipher



Cryptography Techniques

- **Substitution :** استبدال

- The ciphers letters are replaced by other letters

- **Transposition:** النقل

- The ciphers the letters are arranged in a different order

Classical Encryption Techniques

Substitution	Transposition
■ Caesar Cipher ■ Monoalphabetic Cipher ■ Playfair Cipher ■ Hill Cipher ■ Polyalphabetic Ciphers ■ One-Time Pad	• Rail Fence • Row Column Transposition

Substitution Cipher

- ❑ A substitution cipher replaces one symbol with another. If the symbols in the plaintext are alphabetic characters, we replace one character with another.
- ❑ For example, we can replace letter A with letter D, and letter T with letter Z. If the symbols are digits (0 to 9), we can replace 3 with 7, and 2 with 6.

Substitution ciphers types

- Substitution ciphers can be categorized as either
- (1) Monoalphabetic ciphers
- (2) Polyalphabetic ciphers.

Substitution Cipher

- Historical cipher
- Great tool for understanding brute-force vs. analytical attacks
- Encrypts letters rather than bits

Idea: replace each plaintext letter by a fixed other letter.

Plaintext		Ciphertext
A	→	k
B	→	d
C	→	w
....		

for instance, ABBA would be encrypted as kddk

Example(ciphertext):

```
iq ifcc vqqr fb rdq vlllcq na rdq cfjwhwz hr bnnb  hcc  
hwwhbsqvqbre hwq vhlq
```

Substitution Cipher

- Monoalphabetic ciphers

- ✓ A substitution cipher replaces one symbol with another
- ✓ In monoalphabetic substitution, the relationship between a symbol in the plaintext to a symbol in the ciphertext is always one-to-one.

- Polyalphabetic ciphers

- ✓ The relationship between a character in the plaintext and the characters in the ciphertext is one-to-many.

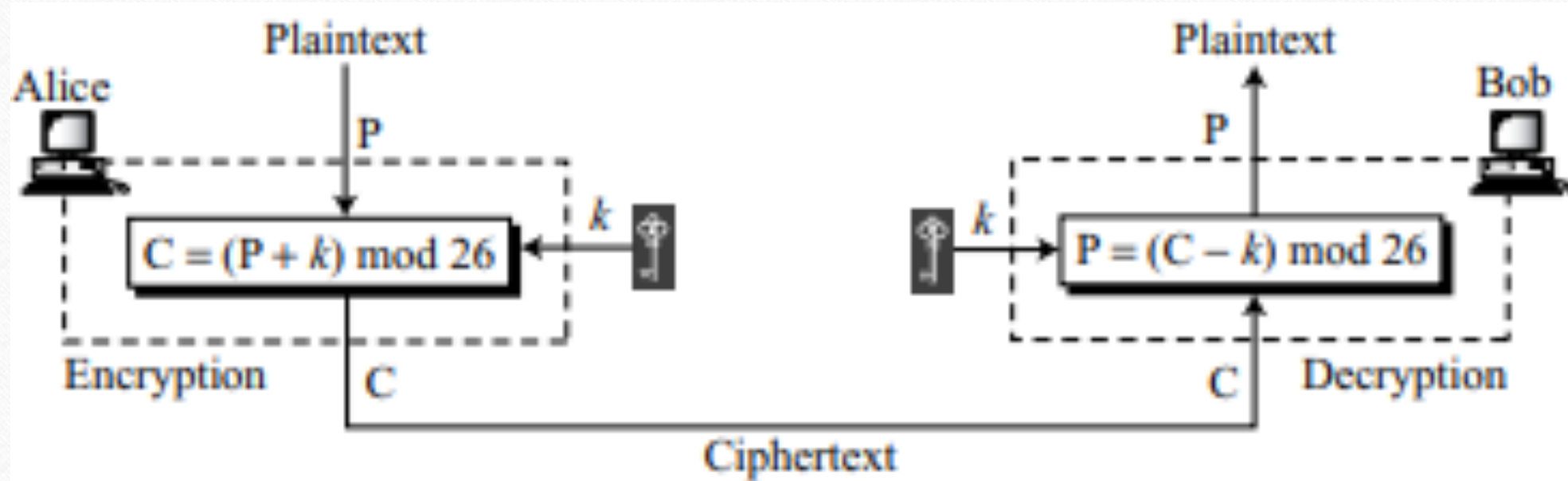
Monoalphabetic ciphers- Examples

- Monoalphabetic ciphers include
 - **Cesar**
 - **Multiplicative**
 - **Affine**
 - **monoalphabetic substitution ciphers.**

[Difference between Monoalphabetic Cipher and Polyalphabetic Cipher - GeeksforGeeks](#)

Different between Monoalphabetic and Polyalphabetic ciphers

Shift (Caesar) Cipher



Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Examples :

Text : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Shift: 23
Cipher: XYZABCDEFGHIJKLMNPOQRSTUVWXYZ

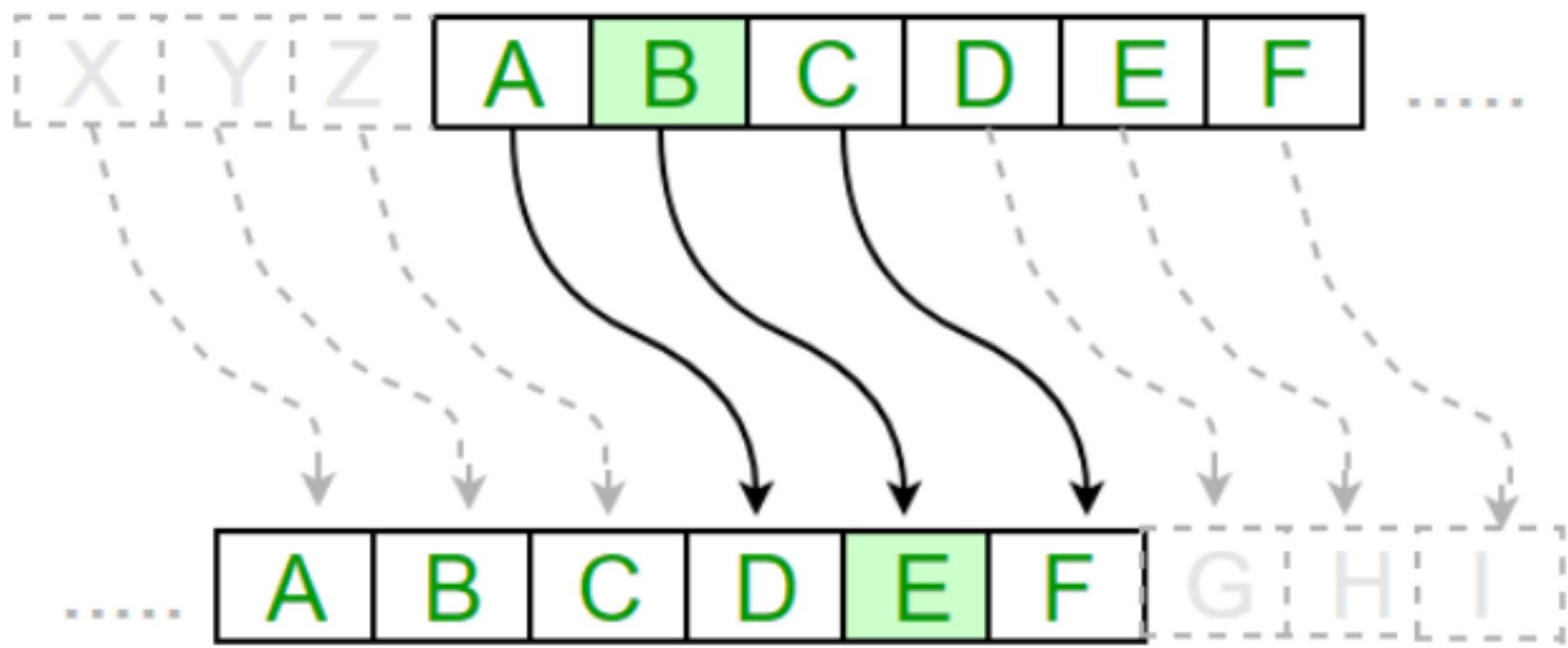
Text : ATTACKATONCE
Shift: 4
Cipher: EXXEGOEXSRGI

$$E_n(x) = (x + n) \bmod 26$$

(Encryption Phase with shift n)

$$D_n(x) = (x - n) \bmod 26$$

(Decryption Phase with shift n)

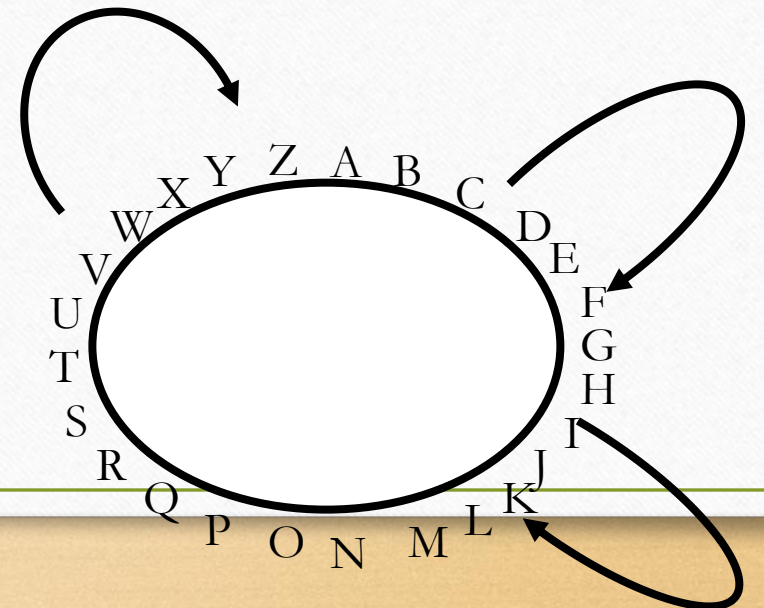


Caesar Cipher

- Letters are replaced by other letters or symbols.
- The earlier known and simplest method used by Julius Caesar
- Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

Caesar Cipher

- Replace each letter of message by a letter a fixed distance away.
- This cipher is sometimes called a shift cipher and sometimes a Caesar cipher.
- Graphically, the Caesar cipher, for $k = 3$ is:



Caesar Cipher:

- can define transformation as:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Example- Caesar Cipher

- Use the Caesar cipher with key = 15 to encrypt the message “hello”.
- **Solution** We apply the encryption algorithm to the plaintext, character by character:

Plaintext: h $\rightarrow$ 07

Plaintext: e $\rightarrow$ 04

Plaintext: l $\rightarrow$ 11

Plaintext: l $\rightarrow$ 11

Plaintext: o $\rightarrow$ 14

Encryption: $(07 + 15) \bmod 26$

Encryption: $(04 + 15) \bmod 26$

Encryption: $(11 + 15) \bmod 26$

Encryption: $(11 + 15) \bmod 26$

Encryption: $(14 + 15) \bmod 26$

Ciphertext: 22 $\rightarrow$ W

Ciphertext: 19 $\rightarrow$ T

Ciphertext: 00 $\rightarrow$ A

Ciphertext: 00 $\rightarrow$ A

Ciphertext: 03 $\rightarrow$ D

Example2:-

- Use the Caesar cipher with key = 15 to decrypt the message “WTAAD”.
- **Solution** We apply the encryption algorithm to the plaintext, character by character:

Ciphertext: W $\rightarrow$ 22

Ciphertext: T $\rightarrow$ 19

Ciphertext: A $\rightarrow$ 00

Ciphertext: A $\rightarrow$ 00

Ciphertext: D $\rightarrow$ 03

Decryption: $(22 - 15) \bmod 26$

Decryption: $(19 - 15) \bmod 26$

Decryption: $(00 - 15) \bmod 26$

Decryption: $(00 - 15) \bmod 26$

Decryption: $(03 - 15) \bmod 26$

Plaintext: 07 $\rightarrow$ h

Plaintext: 04 $\rightarrow$ e

Plaintext: 11 $\rightarrow$ l

Plaintext: 11 $\rightarrow$ l

Plaintext: 14 $\rightarrow$ o

Caesar Cipher- Encryption

- Then we can describe this cipher as:
- Encryption $C = E(p) = (p + 3) \bmod 26$ (shift three spaces)
 - Or
- Encryption $C = E(p) = (p + k) \bmod 26$ (shift k spaces)

Caesar Cipher- Algorithm

- **Input:**

1. Choose a shift value between 1 and 25.
2. Write down the alphabet in order from A to Z.
3. Create a new alphabet by shifting each letter of the original alphabet by the shift value. For example, if the shift value is 3, the new alphabet would be:
4.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
5. Replace each letter of the message with the corresponding letter from the new alphabet. For example, if the shift value is 3, the word “hello” would become “khood”.
6. To decrypt the message, shift each letter back by the same amount. For example, if the shift value is 3, the encrypted message “khood” would become “hello”.

Procedure:

- Traverse the given text one character at a time .
- For each character, transform the given character as per the rule, depending on whether we're encrypting or decrypting the text.
- Return the new string generated.

Advantages:

- Easy to implement and use thus, making suitable for beginners to learn about encryption.
- Can be physically implemented, such as with a set of rotating disks or a set of cards, known as a scytale, which can be useful in certain situations.
- Requires only a small set of pre-shared information.
- Can be modified easily to create a more secure variant, such as by using a multiple shift values or keywords.

Disadvantages:

- It is not secure against modern decryption methods.
- Vulnerable معرض to known-plaintext attacks, where an attacker has access to both the encrypted and unencrypted versions of the same messages.
- The small number of possible keys means that an attacker can easily try all possible keys until the correct one is found, making it vulnerable to a brute force attack.
- It is not suitable for long text encryption as it would be easy to crack كسر.
- It is not suitable for secure communication الاتصال الامن as it is easily broken.
- Does not provide confidentiality السرية, integrity النزاهة, and authenticity المصدقية in a message.

Features of Caesar cipher:

1. Substitution cipher: The Caesar cipher is a type of substitution cipher, where each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet.
2. Fixed key: The Caesar cipher uses a fixed key, which is the number of positions by which the letters are shifted. This key is known to both the sender and the receiver.
3. Symmetric encryption: The Caesar cipher is a symmetric encryption technique, meaning that the same key is used for both encryption and decryption.
4. Limited keyspace: The Caesar cipher has a very limited keyspace of only 26 possible keys, as there are only 26 letters in the English alphabet.
5. Vulnerable to brute force attacks: The Caesar cipher is vulnerable to brute force attacks, as there are only 26 possible keys to try.
6. Easy to implement: The Caesar cipher is very easy to implement and requires only simple arithmetic operations, making it a popular choice for simple encryption tasks.

Brute-force attack

- Try every possible key.
- Until an intelligible translation of the ciphertext into plaintext is obtained.
- Guessing
- Exhaustive key search

Software Tools that can perform brute-force attack

Aircrak-ng	DaveGrah	John the ripper
Cain and abel	Hashcat	Rainbowcrack
Crack	Hydra	Ophcrack

Caesar Cipher Cryptanalysis: brute-force attacks

- It is vulnerable to ciphertext-only attacks using exhaustive key searches (brute-force attacks). The key domain of the additive cipher is very small; there are only 26 keys.
- However, one of the keys, zero, is useless (the ciphertext is the same as the plaintext). This leaves only 25 possible keys. Eve can easily launch a brute force attack on the ciphertext.
- **Example:** Eve has intercepted the ciphertext “UVACLYFZLJBYL”. Show how she can use a brute-force attack to break the cipher.

5 Types of Brute Force Attacks



Simple Brute Force Attacks



Dictionary Attacks



Hybrid Brute Force Attacks



Reverse Brute Force Attacks



Credential Stuffing

Caesar Cipher- Decryption

- Then we can describe this cipher as:
-
- Decryption $P = D(c) = (c - 3) \bmod 26$ (shift three spaces)
- Or
- Decryption $P = D(c) = (c - k) \bmod 26$ (shift k spaces)

Caesar Cipher Cryptanalysis: brute-force attacks

- **Solution:** Eve tries keys from 1 to 7. With a key of 7, the plaintext is “not very secure”, which makes sense.

Ciphertext: UVACLYFZLJBYL

K = 1 → **Plaintext:** tuzbkxeykiaxk

K = 2 → **Plaintext:** styajwdxjhzwj

K = 3 → **Plaintext:** rsxzivcwigyvi

K = 4 → **Plaintext:** qrwyhubvhfxuh

K = 5 → **Plaintext:** pqvxgtaugewtg

K = 6 → **Plaintext:** not very secure

KEY

	PHHW	PH	DIWHU	WKH	WRJD	SDUWB
1	oggv	og	chvgt	vjg	vqic	rctva
2	nffu	nf	bgufs	uif	uphb	qbsuz
3	meet	me	after	the	toga	party
4	ldds	ld	zesdq	sgd	snfz	ozqsx
5	kccr	kc	ydrpc	rfc	rmey	nyprw
6	jbbq	jb	xcqbo	qeb	qldx	mxoqv
7	iaap	ia	wbpan	pda	pkcw	lwnpu
8	hzzo	hz	vaozm	ocz	objv	kvmot
9	gyyn	gy	uznyl	nby	niau	julns
10	fxxm	fx	tymxk	max	mhzt	itkmr
11	ewwl	ew	sxlwj	lzw	lgys	hsjlq
12	dvvk	dv	rwkvi	kyv	kfxr	grikp
13	cuuj	cu	qvjuh	jxu	jewq	fqhjo
14	btti	bt	puitg	iwt	idvp	epgin
15	assh	as	othsf	hvs	hcuo	dofhm
16	zrrg	zr	nsgre	gur	gbtn	cnegl
17	yqqf	yq	mrfqd	ftq	fasm	bmdfk
18	xppe	xp	lqepc	esp	ezrl	alcej
19	wood	wo	kpdob	dro	dyqk	zkbdi
20	vnnc	vn	jocna	cqn	cxpj	yjach
21	ummb	um	inbmz	bpm	bwoi	xizbg
22	tlla	tl	hmaly	aol	avnh	whyaf
23	skkz	sk	glzcx	znk	zumg	vgxze
24	rjjy	rj	fkyjw	ymj	ytlf	ufwyd
25	qiix	qi	ejxiv	xli	xske	tevxc

Example: brute force me

AOPZ PZ TF MPYZA JPWOLY TLZZHNL

THIS IS MY FIRST CIPHER MESSAGE

Caesar Cipher Cryptanalysis: statistical attacks

- Also, It is vulnerable to statistical attacks. This is especially true if the adversary has a long ciphertext. The adversary can use the frequency of occurrence of characters for a particular language.

Table 3.1 *Frequency of occurrence of letters in an English text*

<i>Letter</i>	<i>Frequency</i>	<i>Letter</i>	<i>Frequency</i>	<i>Letter</i>	<i>Frequency</i>	<i>Letter</i>	<i>Frequency</i>
E	12.7	H	6.1	W	2.3	K	0.08
T	9.1	R	6.0	F	2.2	J	0.02
A	8.2	D	4.3	G	2.0	Q	0.01
O	7.5	L	4.0	Y	2.0	X	0.01
I	7.0	C	2.8	P	1.9	Z	0.01
N	6.7	U	2.8	B	1.5		

Example-Caesar Cipher Cryptanalysis: statistical attacks

- **Example:** Eve has intercepted the following ciphertext. Using a statistical attack, find the plaintext.

XLILSYWIMWRS AJSVWEPIJSVJSYVQMPPMSRHSPPEVWMXMWASVX-LQSVILY-
VVCFIJSVIXLIWIPPIVVIGIMZIWQSVISJJIVW

Solution: When Eve tabulates the frequency of letters in this ciphertext, she gets: I =14, V =13, S =12, and so on. The most common character is I with 14 occurrences. This shows that character I in the ciphertext probably corresponds to the character e in plaintext. This means $\text{key} = 4$. Eve deciphers the text to get

the house is now for sale for four million dollars it is worth more hurry before the seller
receives more offers

Exercise 1:

- Decryption A plaintext was encrypted with a Caesar cipher with a shift of 7 (A maps to H).
- The resulting ciphertext is: Kvu'a qbknl h ivvr if paz jvcly
- What was the original plaintext?

Exercise 2:

- Brute Force Use Brute Force to crack the following Caesar ciphertext, to identify the person encoded:
 - KRPCFIJNZWK
- Hint:
- Brute force is best for this question, because there are not enough letters for a frequency analysis (most letters appear once, with only K appearing twice)

Exercise 3:

- Spotting Patterns A plaintext was encrypted with a Caesar cipher, resulting in the following: DOOV ZHOO WKDW HQGV ZHOO
- Can you work out what the plaintext was?
- Hint: Note that there are several occurrences of “OO” appearing within the enciphered words. This must correspond to a double letter in the plaintext. Only some letters in English can be doubled up, so you can try some guesses.

Substitution Cryptography

Techniques

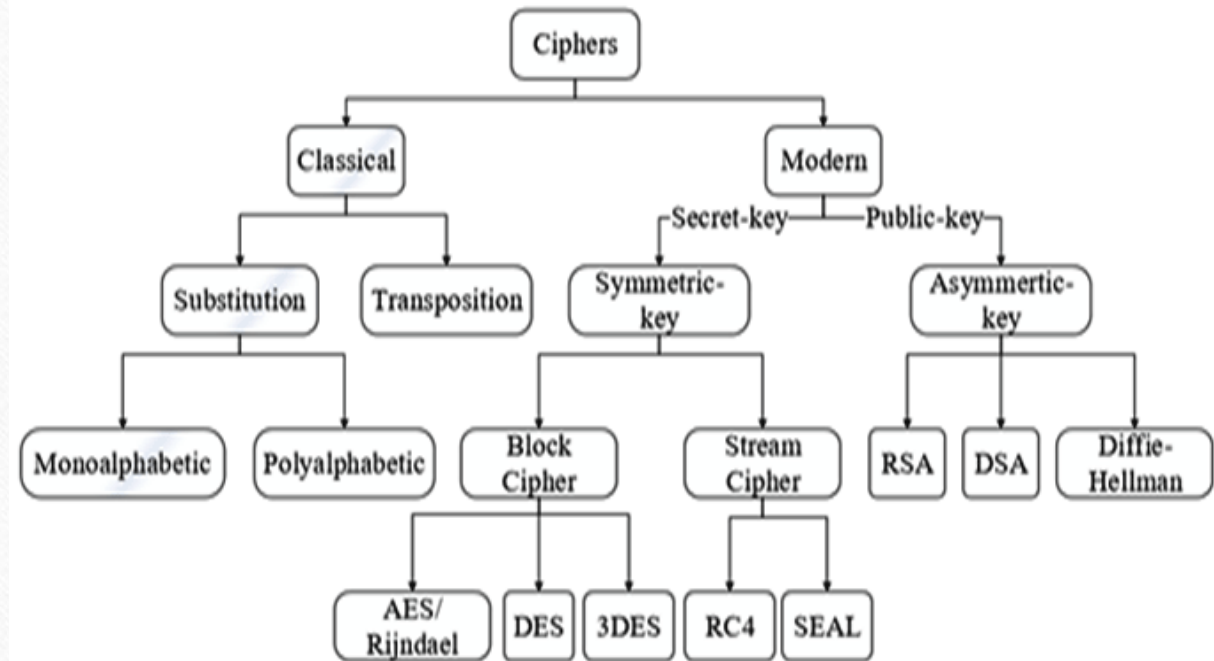
Lecture#3

Contents

- ❑ Classical encryption techniques or Classical cryptosystems.
- ❑ Basics of Playfair cipher and a general comparison of the cipher text with Caesar, Shift, and Monoalphabetic ciphers.
- ❑ Important rules for encryption using Playfair ciphering technique.
- ❑ Use of keyword and construction of 5x5 matrix using the keyword for the encryption process.
- ❑ Creation of digrams and removal of repeating letters in digrams using filler letters.
- ❑ Explanation of cryptanalysis on Monoalphabetic ciphers with example.

Classical Encryption Techniques

- (1) Substitution Technique
- (2) Transposition Technique



Classical Encryption Techniques

Substitution	Transposition
▪ Caesar Cipher ▪ Monoalphabetic Cipher ▪ Polyalphabetic Ciphers ▪ Playfair Cipher ▪ Hill Cipher ▪ One-Time Pad	• Rail Fence • Row Column Transposition

Polyalphabetic Ciphers

- In polyalphabetic substitution, each occurrence of a character may have a different substitute. The relationship between a character in the plaintext to a character in the ciphertext is one-to-many. For example, “a” could be enciphered as “D” in the beginning of the text, but as “N” at the middle. Polyalphabetic ciphers have the advantage of hiding the letter frequency of the underlying language. Eve cannot use single-letter frequency statistic to break the ciphertext.

Playfair Cipher

- Not even the large number of keys in a monoalphabetic cipher provides security
- One approach to improve security was to encrypt multiple letters
- The best-known multiple-letter encryption cipher is the Playfair, which treats digrams in the plaintext as single units and translates these units into ciphertext digrams.
- The Playfair algorithm is based on
 - the use of a 5 * 5 matrix of letters constructed using a keyword.
 - Fill in letters of keyword (sans duplicate بلا تکرار)
 - Fill rest of matrix with other letters.
 - E,g Using keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher

- A polyalphabetic cipher is the Playfair cipher used by the British army during World War I.
- The secret key in this cipher is made of 25 alphabet letters arranged in a 5×5 matrix (letters I and J are considered the same when encrypting).
- Different arrangements of the letters in the matrix can create many different secret keys.

Playfair Cipher

- Aka Playfair square or Wheatstone- Playfair cipher.
- Manual symmetric encryption technique.
- A polyalphabetic cipher
- The first literal digram substitution cipher.
- Invented in 1854 by Charles Wheatstone
- Bore the name of Lord Playfair for promoting its use.

Playfair Cipher

- Multiple letter encryption cipher.
- Digrams
- 5x5 matrix constructed using a keyword (Ex, Monarchy)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Encrypting and Decrypting

- plaintext encrypted two letters at a time:
 1. if a pair is a repeated letter, insert a filler like 'X', eg. "balloon" encrypts as "ba lx lo on"
Diagrams
 2. if both letters fall in the same row, replace each with letter to right (wrapping back to start from end), eg. "ar" encrypts as "RM"
 3. if both letters fall in the same column, replace each with the letter below it (again wrapping to top from bottom), eg. "mu" encrypts to "CM"
 4. otherwise each letter is replaced by the one in its row in the column of the other letter of the pair, eg. "hs" encrypts to "BP", and "ea" to "IM" or "JM" (as desired)

Playfair – rules

(1) Repeating plaintext letters that are in the same pair are separated with a filler letter, such as x, so that balloon would be treated as ba lx lo on

(2) Two plaintext letters that fall in the same row of the matrix are each replaced by the letter to the right, with the first element of the row circularly following the last.

For example, ar is encrypted as RM.

(3) Two plaintext letters that fall in the same column are each replaced by the letter beneath, with the top element of the column circularly following the last. For example, mu is encrypted as CM

(4) Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter. Thus, hs becomes BP and ea becomes IM (or JM, as the encipherer wishes).

Rule for encryption using Playfair Cipher

- 1- Digrams.
- 2- Repeating Letters- filler letter.
- 3- Same column |  | wrap around .
- 4- same row |  | wrap around
- 5- Rectangle |  swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

- Plaintext: attack
- Diagrams: at ta ck
- Plaintext : asks academey
- Diagrams : as ks ac ad em yx
- Plaintext: balloon
- Diagrams: ba ll oo n xxxxx
- Diagrams: ba lx lo on

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example 1

Plaintext : attack

Digrams: at ta ck

at	ta	ck

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Solution- Example 1

at	ta	ck
RS	SR	DE

Plaintext : attack

Cipher text : RSSRDE

Example 2

- Plaintext : mosque
- Digrams: mo sq ue

mo	sq	ue

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Solution – Example 2

mo	sq	ue
ON	TS	ML

Plaintext: mosque

Ciphertext :

ONTSMML

Example 3

- Encrypt the message “Hide the gold under the carpet” using playfair technique with the keyword “Neso Academy”
- Plaintext : Hide the gold under the carpet
- Keyword: Neso Academy
- Solution :
- Ciphertext : IKGDQKDPNRCVEOPQKNDOTVDRZ

Cryptanalysis of a Playfair Cipher

- Obviously a brute-force attack on a Playfair cipher is very difficult.
- The size of the key domain is $25!$ (factorial 25).
- In addition, the encipherment hides the single-letter frequency of the characters.
- However, the frequencies of digrams are preserved (to some extent because of filler insertion), so a cryptanalyst can use a ciphertext-only attack based on the digram frequency test to find the key.

Exercises- playfair

- Encrypt the message “ Hide the image under video” using playfair technique with keyword “computer”

Hill Cipher

- Multi-letter cipher
- Developed by Lester Hill in 1929
- Encrypts a group of letters: digraph, trigraph, or polygraph

Table 3.2 *Grouping of digrams and trigrams based on their frequency in English*

Digram	TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF
Trigram	THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH

Hill Cipher

- A polyalphabetic cipher is the Hill cipher invented by Lester S. Hill. Unlike the other polyalphabetic ciphers, the plaintext is divided into equal-size blocks.
- The blocks are encrypted one at a time in such a way that each character in the block contributes to the encryption of other characters in the block. For this reason, the Hill cipher belongs to a category of ciphers called block ciphers. The other ciphers we studied so far belong to the category called stream ciphers.

Hill Cipher

- In a Hill cipher, the key is a square matrix of size $m \times m$ in which m is the size of the block. If we call the key matrix K , each element of the matrix is $k_{i,j}$ Figure 3.15.

Figure 3.15 *Key in the Hill cipher*

$$K = \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1m} \\ k_{21} & k_{22} & \dots & k_{2m} \\ \vdots & \vdots & & \vdots \\ k_{m1} & k_{m2} & \dots & k_{mm} \end{bmatrix}$$

Let us show how one block of the ciphertext is encrypted. If we call the m characters in the plaintext block $P_1, P_2, \dots, P_m$, the corresponding characters in the ciphertext block are $C_1, C_2, \dots, C_m$. Then we have

$$\begin{aligned}C_1 &= P_1 k_{11} + P_2 k_{21} + \dots + P_m k_{m1} \\C_2 &= P_1 k_{12} + P_2 k_{22} + \dots + P_m k_{m2} \\&\dots \\C_m &= P_1 k_{1m} + P_2 k_{2m} + \dots + P_m k_{mm}\end{aligned}$$

The equations show that each ciphertext character such as C_1 depends on all plaintext characters in the block ($P_1, P_2, \dots, P_m$). However, we should be aware that not all square matrices have multiplicative inverses in $\mathbf{Z}_{26}$, so Alice and Bob should be careful in selecting the key. Bob will not be able to decrypt the ciphertext sent by Alice if the matrix does not have a multiplicative inverse.

Example-Hill cipher

- Using matrices allows Alice to encrypt the whole plaintext. In this case, the plaintext is an $l \times m$ matrix in which l is the number of blocks. For example, the plaintext “code is ready” can make a 3×4 matrix when adding extra bogus character “z” to the last block and removing the spaces.
- The ciphertext is “OHKNIHGKLISS”. Bob can decrypt the message using the inverse of the key matrix. Encryption and decryption are shown in Figure 3.16.

Figure 3.16 *Example 3.20*

$$\begin{array}{c} \mathbf{C} \\ \left[\begin{array}{cccc} 14 & 07 & 10 & 13 \\ 08 & 07 & 06 & 11 \\ 11 & 08 & 18 & 18 \end{array} \right] \end{array} = \begin{array}{c} \mathbf{P} \\ \left[\begin{array}{cccc} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{array} \right] \end{array} \begin{array}{c} \mathbf{K} \\ \left[\begin{array}{cccc} 09 & 07 & 11 & 13 \\ 04 & 07 & 05 & 06 \\ 02 & 21 & 14 & 09 \\ 03 & 23 & 21 & 08 \end{array} \right] \end{array}$$

a. Encryption

$$\begin{array}{c} \mathbf{P} \\ \left[\begin{array}{cccc} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{array} \right] \end{array} = \begin{array}{c} \mathbf{C} \\ \left[\begin{array}{cccc} 14 & 07 & 10 & 13 \\ 08 & 07 & 06 & 11 \\ 11 & 08 & 18 & 18 \end{array} \right] \end{array} \begin{array}{c} \mathbf{K}^{-1} \\ \left[\begin{array}{cccc} 02 & 15 & 22 & 03 \\ 15 & 00 & 19 & 03 \\ 09 & 09 & 03 & 11 \\ 17 & 00 & 04 & 07 \end{array} \right] \end{array}$$

b. Decryption

The Hill Algorithm

- This can be expressed as
- $C = E(K, P) = P \times K \bmod 26$
- $P = D(K, C) = CK^{-1} \bmod 26 = P \times K \times K^{-1} \bmod 26$

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$


$$\begin{aligned} C_1 &= (P_1 K_{11} + P_2 K_{21} + P_3 K_{31}) \bmod 26 \\ C_2 &= (P_1 K_{12} + P_2 K_{22} + P_3 K_{32}) \bmod 26 \\ C_3 &= (P_1 K_{13} + P_2 K_{23} + P_3 K_{33}) \bmod 26 \end{aligned}$$

Hill Cipher Example

Encrypt “pay more money” using Hill cipher with key

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

Key = 3x3 matrix

Plaintext = pay mor emo ney

Hill Cipher solution

Encryption : pay

p	q	y	m	o	r	e	m	o	n	e	y
15	0	24	12	14	17	4	12	14	13	4	24

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \mod 26$$

$$(C_1 \ C_2 \ C_3) = (15 \ 0 \ 24) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \mod 26$$

$$= (15 \times 17 + 0 \times 21 + 24 \times 2 \quad 15 \times 17 + 0 \times 18 + 24 \times 2 \quad 15 \times 5 + 0 \times 21 + 24 \times 19) \mod 26$$

$$= (303 \ 303 \ 531) \mod 26$$

$$= (17 \ 17 \ 11)$$

R R L

Encrypting : mor

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{mod } 26$$

$$(C_1 \ C_2 \ C_3) = (12 \ 14 \ 17) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{mod } 26$$

$$= (12 \times 17 + 14 \times 21 + 17 \times 2 \quad 12 \times 17 + 14 \times 18 + 17 \times 2 \quad 12 \times 5 + 14 \times 21 + 17 \times 19) \text{mod } 26$$

$$= (532 \ 490 \ 677) \text{mod } 26$$

$$= (12 \ 22 \ 1)$$

$$= (\underline{M} \ W \ p)$$

Encrypting: ney

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \text{mod } 26$$

$$(C_1 \ C_2 \ C_3) = (13 \ 4 \ 24) \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \text{mod } 26$$

$$= (13 \times 17 + 4 \times 21 + 24 \times 2 \quad 13 \times 17 + 4 \times 18 + 24 \times 2 \quad 13 \times 5 + 4 \times 21 + 24 \times 19) \text{mod } 26$$

$$= (348 \ 312 \ 538) \text{mod } 26$$

= (P D H)

Plaintext : pay more money

Ciphertext : RRLMWBKASPDH

Decryption Hill Cipher

- Decryption requires K^{-1} , the inverse matrix K

$$K^{-1} = \frac{1}{\text{Det } K} \times \text{Adj } K$$

To find Det K , adj K

Decryption cipher

To find the determinant of K :

$$\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$$

$$\text{Det} \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$\begin{aligned} &= 17(18 \times 19 - 2 \times 21) - 17(19 \times 21 - 2 \times 21) + 5(2 \times 21 - 2 \times 18) \bmod 26 \\ &= 17(342 - 42) - 17(399 - 42) + 5(42 - 36) \bmod 26 \\ &= 17(300) - 17(357) + 5(6) \bmod 26 \end{aligned}$$

$$\text{Det} \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix} \bmod 26$$

$$\begin{aligned} &= 17(18 \times 19 - 2 \times 21) - 17(19 \times 21 - 2 \times 21) + 5(2 \times 21 - 2 \times 18) \bmod 26 \\ &= 17(342 - 42) - 17(399 - 42) + 5(42 - 36) \bmod 26 \\ &= 17(300) - 17(357) + 5(6) \bmod 26 \\ &= 5100 - 6069 + 30 \bmod 26 \\ &= -939 \bmod 26 \\ &= -3 \bmod 26 \end{aligned}$$

$$= 23$$

Hill cipher

- Hill cipher is a polygraphic substitution cipher based on linear algebra.
- Each letter is represented by a number modulo 26. Often the simple scheme A = 0, B = 1, ..., Z = 25 is used, but this is not an essential feature of the cipher.
- To encrypt a message, each block of n letters (considered as an n -component vector) is multiplied by an invertible $n \times n$ matrix, against modulus 26.
- To decrypt the message, each block is multiplied by the inverse of the matrix used for encryption.
- The matrix used for encryption is the cipher key, and it should be chosen randomly from the set of invertible $n \times n$ matrices (modulo 26).

Advantages of Hill Cipher

- Hill cipher offers several advantages as an encryption algorithm:
- **Security:** Hill cipher provides a higher level of security compared to traditional substitution ciphers as it operates on blocks of letters rather than individual letters. This makes it more resistant to frequency analysis attacks.
- **Flexibility:** Hill cipher allows encryption and decryption of messages containing both uppercase and lowercase letters, as well as punctuation marks and spaces. This flexibility makes it suitable for encrypting a wide range of text-based data.
- **Complexity:** The encryption process of Hill cipher involves matrix operations, making it more complex and harder to decipher without knowledge of the encryption key. This adds an additional layer of security to the algorithm.

Advantages of Hill Cipher, cont.,

- **Key Strength:** The security of Hill cipher is directly dependent on the size and randomness of the encryption key matrix. By using a larger key matrix, the encryption strength can be increased, making it more difficult for unauthorized individuals to decrypt the message.
- **Mathematical Background:** Hill cipher is based on linear algebra principles, providing a foundation for learning and understanding more advanced encryption techniques. It offers an opportunity to explore the relationship between matrices and encryption algorithms.

Contents

- Definition Polyalphabetic substitution Cipher
- VIGENÈRE CIPHER
- VERNAM CIPHER
- One Time Pad
- Rail Fence Technique

Polyalphabetic Cipher

- To improve on the simple monoalphabetic technique.
- General name : Polyalphabetic Substitution Cipher.
- The feature of Polyalphabetic Cipher :-
 - A set of related monoalphabetic substitution rules is used.
 - A key determines which particular rule is chosen for a given transformation.

VIGENÈRE CIPHER

- The best known, and one of the simplest, **polyalphabetic ciphers** is the Vigenère cipher.
- In this scheme, the set of related monoalphabetic substitution rules consists of the 26 Caesar ciphers with shifts of 0 through 25.
- Each cipher is denoted by a key letter, which is the ciphertext letter that substitutes for the plaintext letter a.
- Thus, a Caesar cipher with a shift of 3 is denoted by the key value 3

VIGENÈRE CIPHER

- We can express the Vigenère cipher in the following manner.
- Assume a sequence of plaintext letters $P = p_0, p_1, p_2, \dots, p_{n-1}$ and a key consisting of the sequence of letters $K = k_0, k_1, k_2, \dots, k_{m-1}$, where typically $m < n$.
- The sequence of ciphertext letters $C = C_0, C_1, C_2, \dots, C_{n-1}$ is calculated as

$$\begin{aligned} C = C_0, C_1, C_2, \dots, C_{n-1} &= E(K, P) = E[(k_0, k_1, k_2, \dots, k_{m-1}), (p_0, p_1, p_2, \dots, p_{n-1})] \\ &= (p_0 + k_0) \bmod 26, (p_1 + k_1) \bmod 26, \dots, (p_{m-1} + k_{m-1}) \bmod 26, \\ &\quad (p_m + k_0) \bmod 26, (p_{m+1} + k_1) \bmod 26, \dots, (p_{2m-1} + k_{m-1}) \bmod 26, \dots \end{aligned}$$

VIGENÈRE CIPHER

- Thus, **the first letter of the key is added to the first letter of the plaintext, mod 26**, the second letters are added, and so on through the first m letters of the plaintext. For the next m letters of the plaintext, the key letters are repeated. This process continues until all of the plaintext sequence is encrypted. A general equation of the encryption process is

General Rule- VIGENÈRE cipher

It consists of the 26 Caesar ciphers with shifts of 0 through 25

To encrypt a message, a key is needed that is as long as the message.

Encryption Process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Decryption Process

$$P_i = (C_i - K_{i \bmod m}) \bmod 26$$

Example

- If the keyword is **deceptive**, the message “we are discovered save yourself” is encrypted

To encrypt a message, a key is needed that is as long as the message.

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Encryption Process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Example- VIGENÈRE cipher

- Key : deceptivedeceptivedeceptive
- Plaintext : wearediscoveredsaveyourself
- Ciphertext : ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17

Key	15	19	8	21	4	3	4	2	4	15	19	8	21	4
PT	4	3	18	0	21	4	24	14	20	17	18	4	11	5
CT	19	22	0	21	25	7	2	16	24	6	11	12	6	9

VIGENÈRE Cipher Cryptanalysis

- Problem :
- An analyst looking at only the ciphertext would detect the repeated sequences **VTW** at a displacement of 9 and make the assumption that the keyword is either three or nine letters in length.
- The appearance of **VTW** twice could be by chance and may not reflect identical plaintext letters encrypted with identical key letters.
- However, if the message is long enough, there will be a number of such repeated ciphertext sequences. By looking for common factors in the displacements of the various sequences, the analyst should be able to make a good guess of **the keyword length**
- Determining the length of the keyword.
- Key and the plaintext share **the same frequency distribution of letters**, a statistical technique can be applied

VIGENÈRE Cipher Cryptanalysis

- Solution of the cipher now depends on an important insight.
- If the keyword length is m , then the cipher, in effect, consists of m monoalphabetic substitution ciphers.
- For example, with the keyword DECEPTIVE, the letters in positions 1, 10, 19, and so on are all encrypted with the same monoalphabetic cipher. Thus, we can use the known frequency characteristics of the plaintext language to attack each of the monoalphabetic ciphers separately

Autokey system

- The periodic nature of the keyword can be eliminated by using a non-repeating keyword that is as long as the message itself.
- VIGENÈRE proposed **autokey system**, in which a keyword is concatenated with the plaintext itself to provide a running key.
- Example :
- Key : deceptivewearediscoveredsav
- Plaintext : wearediscoveredsaveyourself
- Ciphertext : ZICVTWQNGKZEIIGASXSTSLVWLA

Autokey system

- Example :
- Key : deceptivewarediscoveredsav
- Plaintext : wearediscoveredsaveyourself
- Ciphertext : ZICVTWQNGKZEIIGASXSTSLVWLA

VERNAM CIPHER

- **It works on binary data (bits) rather than letters.**
- Need of ultimate defense against cryptanalytic attack.
- Length of the keyword =Length of the Plaintext.
- No statistical relationship to it .
- AT& T engineer named Gilbert Vernam in 1918.
- His system works **on binary bits rather than letter.**
- The system can be expressed as follows:-

VERNAM CIPHER

- His system works on binary data (bits) rather than letters. The system can be expressed succinctly as follows (Figure 3.7):

$$c_i = p_i \oplus k_i$$

where

p_i = i th binary digit of plaintext

k_i = i th binary digit of key

c_i = i th binary digit of ciphertext

$\oplus$ = exclusive-or (XOR) operation



Figure 1: XOR gate, with inputs A and B and outputs Q

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

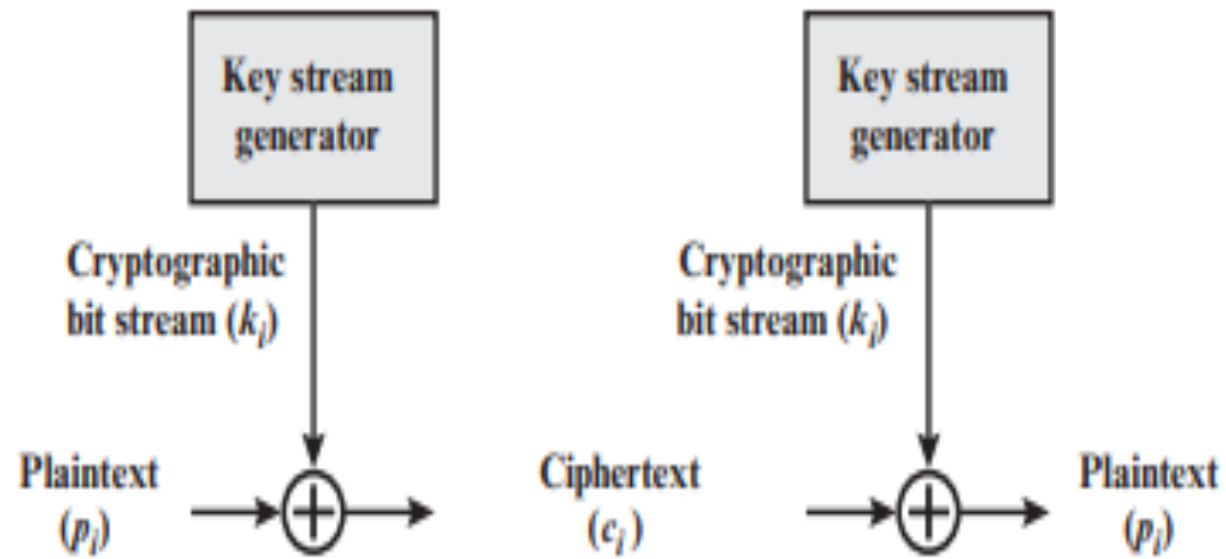
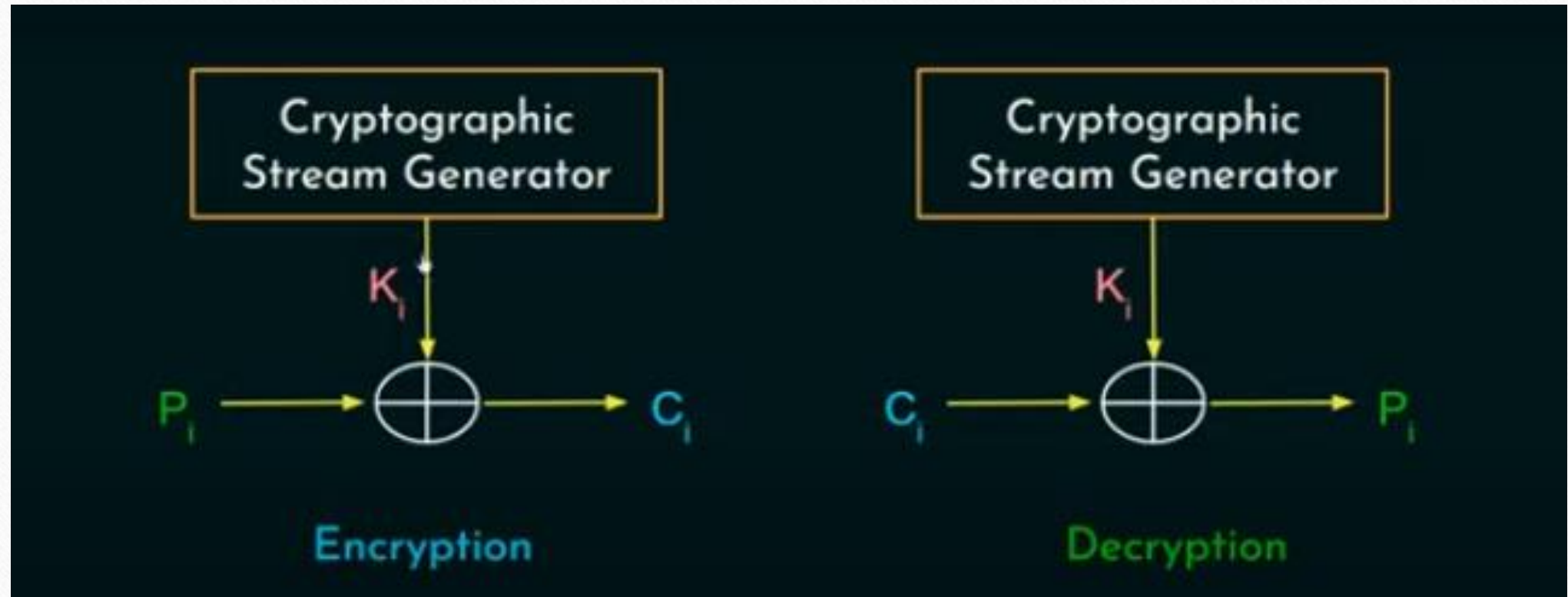


Figure 3.7 Vernam Cipher

Vernam Cipher



Example

- Encrypt the plaintext “01001000” using VERNAM CIPHER where the key is “01010000”

- **Plaintext :** 01001000
- **Key :** 01010000
- **Ciphertext :** 00011000

VERNAM CIPHER

- Compare this with Equation (3.3) for the Vigenère cipher. Thus, the ciphertext is generated by performing the bitwise XOR of the plaintext and the key. Because of the properties of the XOR, decryption simply involves the same bitwise operation:

$$p_i = c_i \oplus k_i$$

- which compares with Equation (3.4).

Vernam Cipher- Cryptanalysis

- The essence of this technique is the means of construction of the key.
- Vernam proposed the use of a running loop of tape that eventually repeated the key, so that in fact the system worked with a very long but repeating keyword.
- Although such a scheme, with a long key, presents formidable cryptanalytic difficulties, it can be broken with sufficient ciphertext, the use of known or probable plaintext sequences, or both

One-Time Pad

- An Army Signal Corp officer, Joseph Mauborgne, proposed an improvement to the Vernam cipher that yields the ultimate in security.
- Mauborgne suggested using a random key that is as long as the message, so that the key need not be repeated.
- In addition, the key is to be used to encrypt and decrypt a single message, and then is discarded.
- Each new message requires a new key of the same length as the new message.
- Such a scheme, known as a one-time pad, is unbreakable.
- It produces random output that bears no statistical relationship to the plaintext. Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the codes

One-Time Pad

- Improvement to the Vernam cipher.
- It yield the ultimate in security.
- Random key that is as long as the message.
- The key need not be repeated.
- In addition, the key is to be used to encrypt and decrypt a single message and then is discarded.
- Each new message requires a new key of the same length as the new message.
- Such a scheme , known as a one-time pad is unbreakable.

One Time Pad

- It produces random output.
- No statistical relationship to the plaintext.
- Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code.
- The code is unbreakable.
- The security of the one-time pad is entirely due to the randomness of the key.

One-time pad (unbreakable cipher)

- It is **Vigener cipher** with **key length equal to the length of Plaintext**. Also the key must be chosen in **a random way and can only be used once**. It produces random output that bears no statistical relationship to the plain text. The one-Time pad is only cryptosystem that exhibits what is referred to as perfect secrecy.

$$C = E(P) = (P + ki) \bmod n$$

$$P = D(C) = (C - ki) \bmod n$$

Example- One-Time Pad

- Encrypt the plaintext “ attack now” using one-time pad cipher where the key is “ Kjuimg rtn”

Index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z

$$C = E(P) = (P + ki) \bmod \underline{n=26}$$

Key : Kjuimg rtn

Plaintext : attack now

Ciphertext : kcnioq ehi

Example

- Suppose that we are using a Vigenère scheme with 27 characters in which the twenty-seventh character is the space character, but with a one-time key that is as long as the message.
- Consider the ciphertext

ANKYODKYUREPFJBYOJDSPLEIYIUNOFDOIUERFPLUYTS

- We now show two different decryptions using two different keys:

ciphertext: ANKYODKYUREPFJBYOJDSPREYIUNOFDOIUERFPLUYTS
key: *pxlmvmsydofuyrvzwc tnlebnecvgdupahfzzlmnyih*
plaintext: mr mustard with the candlestick in the hall

ciphertext: ANKYODKYUREPFJBYOJDSPREYIUNOFDOIUERFPLUYTS
key: *pftgpmiydgaxgoufhkl11lmhsqdgogtewbqfgyovuhwt*
plaintext: miss scarlet with the knife in the library

Two Fundamentals Difficulties- One-Time Pad

- In theory, we need look no further for a cipher. **The one-time pad offers complete security but, in practice, has two fundamental difficulties:**
 - (1) **There is the practical problem of making large quantities of random keys.** Any heavily used system might require millions of random characters on a regular basis. Supplying truly random characters in this volume is a significant task.
 - (2) **Even more daunting is the problem of key distribution and protection.**
 - For every message to be sent, a key of equal length is needed by both sender and receiver. Thus, a mammoth key distribution problem exists.
 - Because of these difficulties, the one-time pad is of limited utility and is useful primarily for low bandwidth channels requiring very high security.
 - The one-time pad is the only cryptosystem that exhibits what is referred to as perfect secrecy

Perfect Secrecy

- The one-time pad is the cryptosystem that exhibits what is referred to as perfect secrecy.
- Perfect secrecy is the notion that, given an encrypted message (or ciphertext) from a perfectly secure encryption system (or cipher), absolutely nothing will be revealed about the unencrypted message (or plaintext) by the ciphertext.

Transposition Techniques

- A very different kind of mapping is achieved by performing some sort of permutation on the plaintext letters.
- This technique is referred to as a transposition cipher.
- The simplest such cipher is Rail Fence.

Rail Fence- transposition Cipher

- The plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.

Example1 – Rail Fence

- To encipher the message
- Plain text : “meet me after the toga party”
- Depth: 2

m		e		m		a		t		r		h		t		g		p		r		y
	e		t		e		f		e		t		e		o		a		a		t	

- Ciphertext : MEMATRHTGPRYETEFETEOAAT

Example- Rail Fence

- Plaintext : neso academy is the best
- Depth : 2

n		s		a		a		e		y		s		h		b		s	
	e		o		c		d		m		i		t		e		e		t

- Ciphertext: NSAAEYSHBSEOCDMITEET

Example

- Plaintext :Thank you very much
- Depth :3

T				k				v				m			
	h		n		y		u		e		y		u		h
		a				o				r				c	

- Ciphertext :TKVMHNYUEYUHAORE

Row column transposition

- A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns. The order of the columns then becomes the key to the algorithm. F
- A more complex scheme
- Rectangle
- Write : row by row
- Read : column by Column
- Key : order of the column

Example1

Key:	4 3 1 2 5 6 7
Plaintext:	a t t a c k p o s t p o n e d u n t i l t w o a m x y z
Ciphertext:	TTNAAPTMTSUOAODWCOIXKNLYPETZ

Example 1

- Thus, in this example, the key is 4312567.
- To encrypt, start with the column that is labeled 1, in this case column 3. Write down all the letters in that column. Proceed to column 4, which is labeled 2, then column 2, then column 1, then columns 5, 6, and 7.

Cryptanalysis- transposition

- A pure transposition cipher is easily recognized because it has the same letter frequencies as the original plaintext. For the type of columnar transposition just shown, cryptanalysis is fairly straightforward and involves laying out the ciphertext in a matrix and playing around with column positions. Digram and trigram frequency tables can be useful.
- The transposition cipher can be made significantly more secure by performing more than one stage of transposition.
- The result is a more complex permutation that is not easily reconstructed. Thus, if the foregoing message is reencrypted using the same algorithm

```

Key:      4 3 1 2 5 6 7
Input:    t t n a a p t
          m t s u o a o
          d w c o i x k
          n l y p e t z
Output:   NSCYAUOPTTWLTMDNAOIEPAXTTOKZ

```

To visualize the result of this double transposition, designate the letters in the original plaintext message by the numbers designating their position. Thus, with 28 letters in the message, the original sequence of letters is

```

01 02 03 04 05 06 07 08 09 10 11 12 13 14
15 16 17 18 19 20 21 22 23 24 25 26 27 28

```

After the first transposition, we have

```

03 10 17 24 04 11 18 25 02 09 16 23 01 08
15 22 05 12 19 26 06 13 20 27 07 14 21 28

```

which has a somewhat regular structure. But after the second transposition, we have

```

17 09 05 27 24 16 12 07 10 02 22 20 03 25
15 13 04 23 19 14 11 01 26 21 18 08 06 28

```

This is a much less structured permutation and is much more difficult to cryptanalyze.

Example 2:

- Plaintext : “ Kill Corona Virus at twelve am tomorrow”
- Key : 4 3 1 2 5 6 7

Key → 4 3 1 2 5 6 7

Plaintext (Input) →

K	i	l	l	c	o	r
o	n	a	v	i	r	u
s	a	t	t	w	e	l
v	e	a	m	t	o	m
o	r	r	o	w	y	z

- Ciphertext : LATARLV'TMOINAERKOSVOCIW'TWOREOYRULM

Example 2– More complex

Plaintext : “ Kill Corona Virus at twelve am tomorrow”

Key : 4 3 1 2 5 6 7



The diagram shows a 5x7 grid of letters. Above the grid, the key sequence 4 3 1 2 5 6 7 is shown with an orange arrow pointing to the first row. To the left of the grid, the label 'Ciphertext (Input)' is shown with a blue arrow pointing to the first column. A red curved arrow points from the bottom-left of the grid back towards the 'Ciphertext (Input)' label.

Key →	4	3	1	2	5	6	7
Ciphertext (Input) →	L	A	T	A	R	L	V
	T	M	O	I	N	A	E
	R	K	O	S	V	O	C
	I	W	T	W	O	R	E
	O	Y	R	U	L	M	Z

Ciphertext : TOOTRAISWUAMKWYLTRIORNVOLLAORMVECEZ

Homework

- (1) Decrypt the ciphertext “dfhhri zom Isek” using one-time pad cipher where the random key is “abcdef ghi jklm”
- (2) Decrypt the ciphertext “0001110000011001” using VERNAM cipher and the key is “ 00101101”
- (3) Decrypt the ciphertext “dyzien” with key “3 5 6 8 2” using Vigenere cipher